

BODY PERFORMANCE ANALYTICS

COMPREHENSIVE AI & ML LEARNING
PROJECT REPORT (TQV FRAMEWORK)

PRESENTED TO

Prof. Sherif Morsy

PRESENTED BY

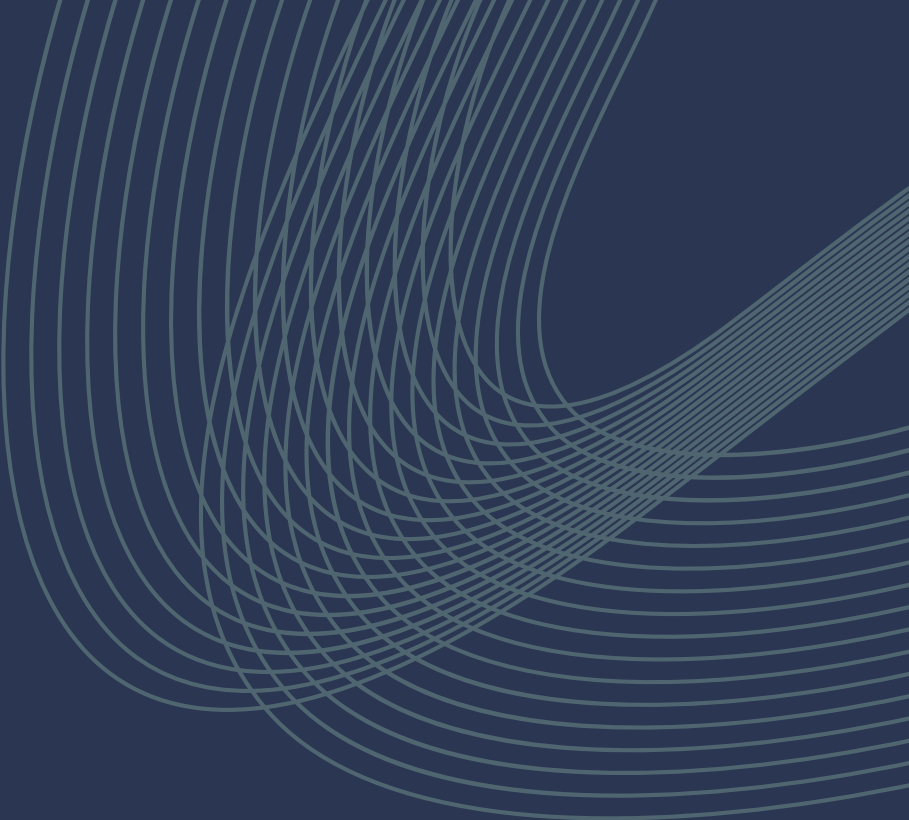
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Body Performance Analytics

COURSE: INTRODUCTION TO AI AND ML

PROJECT: BODY PERFORMANCE
CLASSIFICATION AND REGRESSION

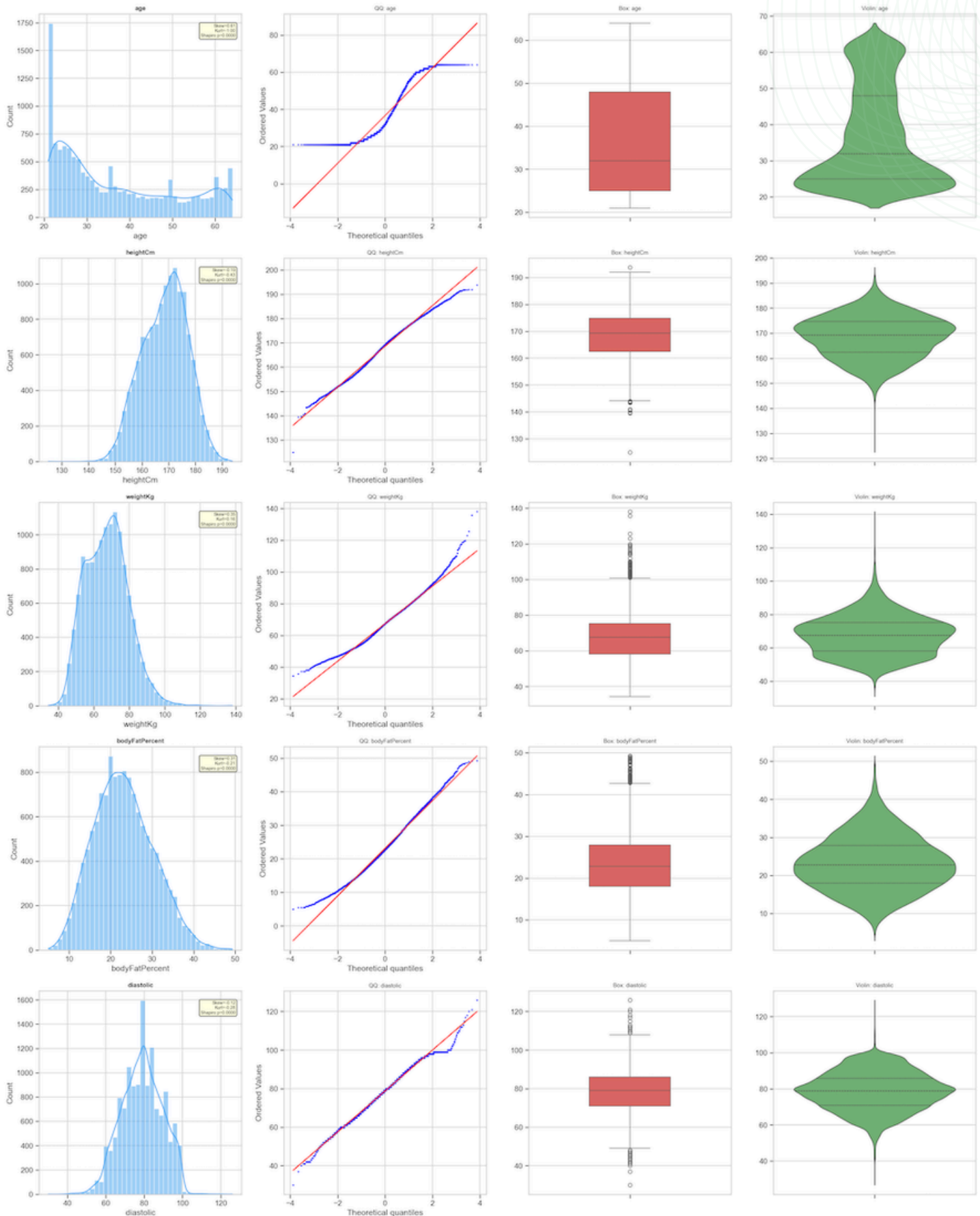
DATASET: BODY PERFORMANCE (KAGGLE)
— 13,393 ROWS × 12 COLUMNS

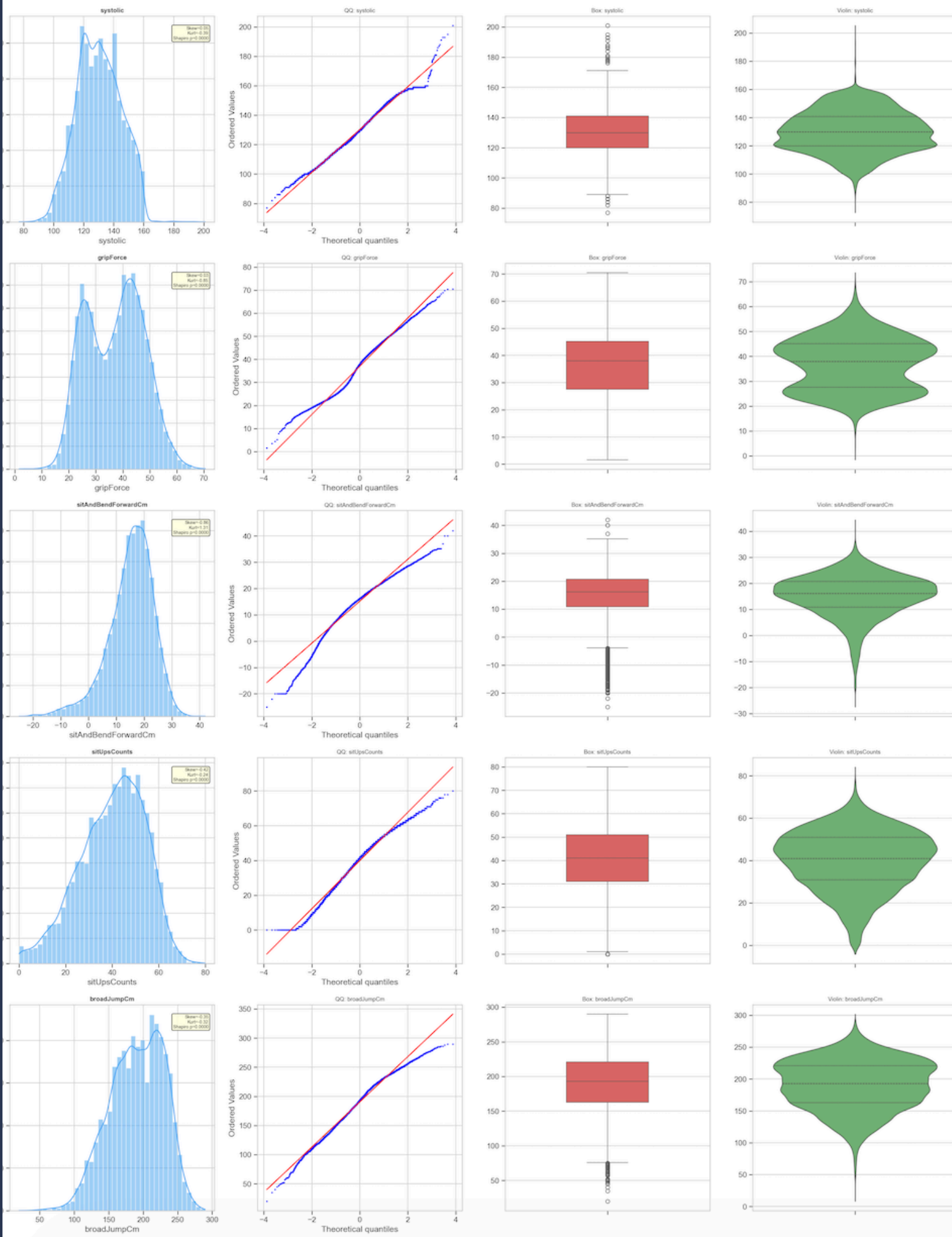
Part 1: Data Preparation & Exploratory Data Analysis

1. Dataset Overview

The dataset contains 13,393 fitness test records from a national fitness program. Each row represents one person's measurements: age, gender, height, weight, body fat %, blood pressure (systolic/diastolic), grip force, flexibility (sit-and-bend), endurance (sit-ups), explosive power (broad jump), and a performance class (A/B/C/D).

Distribution analysis of all variables





2. Column Understanding

Demographics & Metrics

Column	Type	Constraints	Description
age	Numeric	21–64	Age in years
gender	Categorical	M or F	Biological sex
heightCm	Numeric	100–220 cm	Height in cm
weightKg	Numeric	20–200 kg	Body weight
bodyFatPercent	Numeric	3–60%	Body fat percentage

Vitals & Performance

Column	Type	Constraints	Description
diastolic	Numeric	< systolic	Resting BP (mmHg)
systolic	Numeric	> diastolic	Contraction BP (mmHg)
gripForce	Numeric	Positive	Grip strength (kg)
sitAndBendForwardCm	Numeric	-30 to +50	Flexibility (cm past toes)
sitUpsCounts	Numeric	Non-negative	Sit-ups count (core)
broadJumpCm	Numeric	Positive	Broad jump distance (cm)
performanceClass	Categorical	A, B, C, D	A = best, D = worst

3. Data Type Verification

All columns have correct data types. Numeric features are stored as float64, gender and class as object (string). No type conversion was needed.

4. Missing Values Analysis

Result: Zero missing values across all 12 columns.

No imputation techniques were required. This was verified by computing null counts and percentages for every column.

5. Duplicate Detection

Duplicate rows were found, but the decision was made to keep them. In fitness testing without participant IDs, two different people CAN have identical measurements. Without additional metadata, we cannot distinguish true duplicates from coincidental matches.

6. Data Validity Checks & Cleaning

Fix 1: Diastolic $\geq$ Systolic (5 rows dropped)

How we found it: Medical science dictates that systolic pressure (heart contraction) must always exceed diastolic (heart relaxation). We filtered for rows where `diastolic  $\geq$  systolic`.

Why we removed them: These values represent measurement errors or data-entry mistakes — no living human can have diastolic $\geq$ systolic.

Fix 2: Extreme Body Fat Percentages (rows dropped)

How we found it: We checked physiological limits — essential body fat is 3–5% for men, 10–13% for women. Values above 50% represent extreme cases that are likely errors.

Action: Dropped rows where body fat ` $< 5\%$ `, ` $> 50\%$ `, female ` $< 10\%$ `, or male ` $> 45\%$ `.

Fix 3: Zero Values (rows dropped)

How we found it: Zero grip force (0 kg), zero broad jump (0 cm), and zero blood pressure (0/0 mmHg) are physically impossible for a conscious person performing a test.

- Grip = 0: Even severe sarcopenia produces 5–10 kg of grip. Zero means equipment failure.
- Broad jump = 0: Zero distance means the test was not performed (missing data as zero).
- BP = 0/0: Clinically dead — clearly missing data.
- Sit-ups = 0 & age < 60: Unrealistic for adults under 60 → dropped.
- Sit-ups = 0 & age ≥ 60: Kept — age-related sarcopenia makes this physiologically valid.

Fix 4: Sit-and-Bend Range [-30, +50] cm

How we found it: Max values included 213 cm and 185 cm — no human bends forward 2+ meters.

Fix 5: Diastolic < 20 mmHg (3 rows dropped)

How we found it: Below 20 mmHg is incompatible with consciousness — measurement error.

Fix 6: BMI < 14

How we found it: We computed $BMI = \text{weight}/(\text{height}/100)^2$ and filtered for values below 14. BMI < 14 is incompatible with life.

Fix 7: Broad Jump > 290 cm (16 rows dropped)

How we found it: Standing broad jumps above 290 cm are Olympic-caliber — impossible in a general population sample.

Column Renaming

Why: Original names contained spaces, hyphens, and special characters (`body fat\_%`, `sit-ups counts`). We standardized all column names to camelCase for code compatibility.

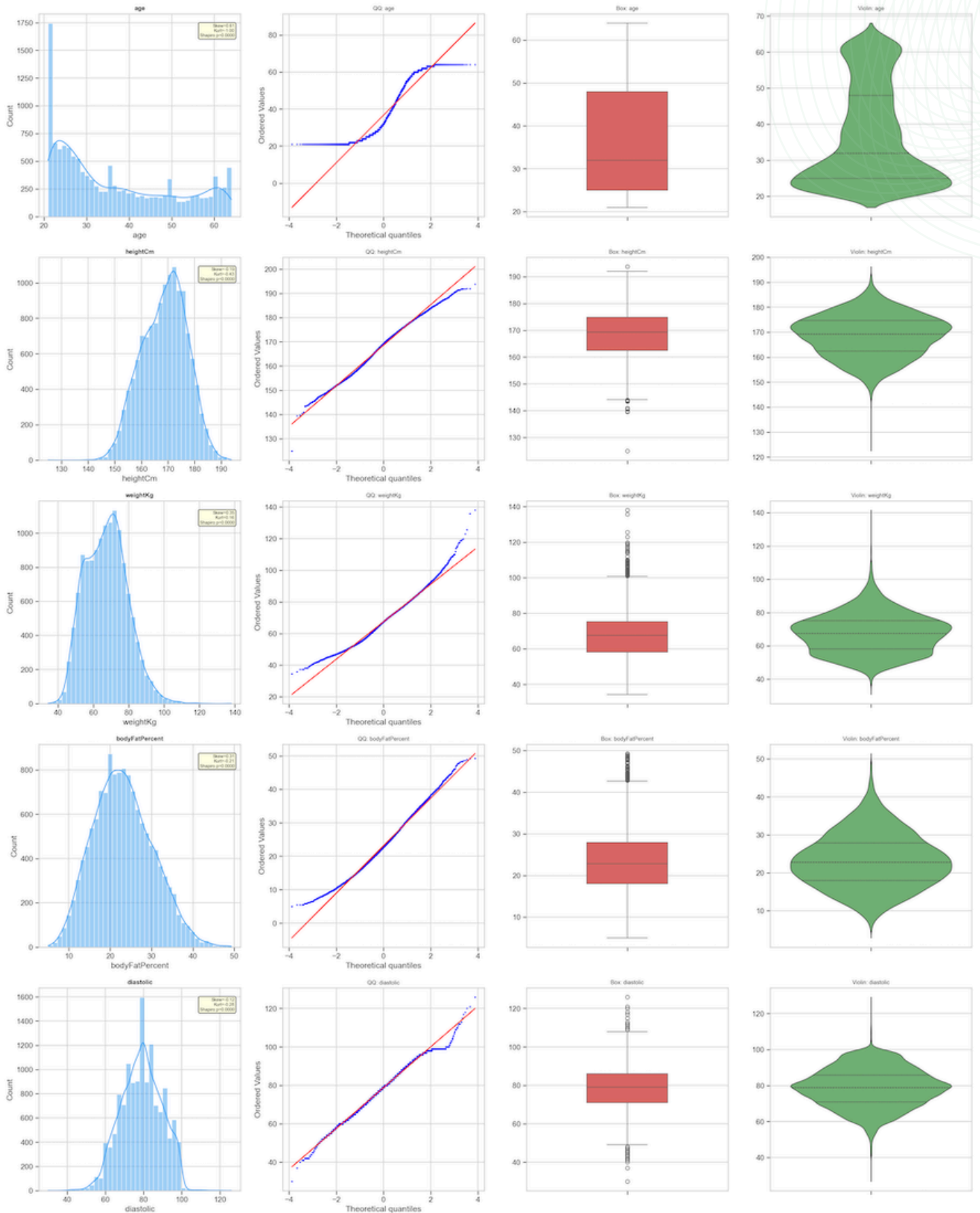
Cleaning Summary

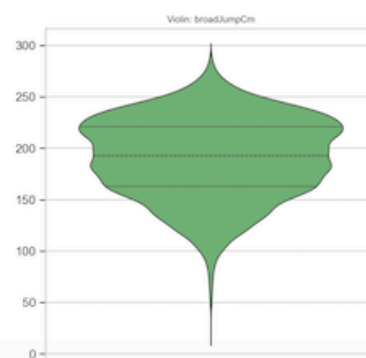
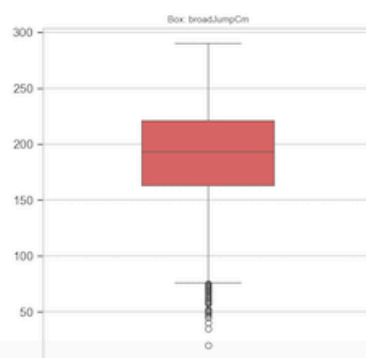
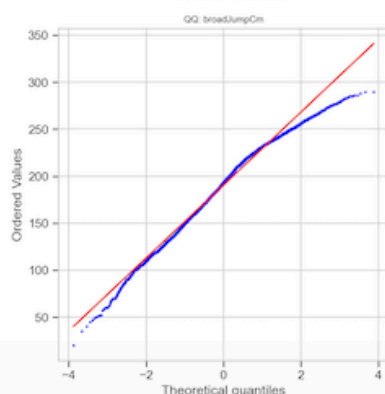
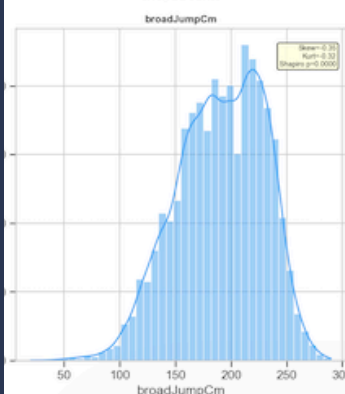
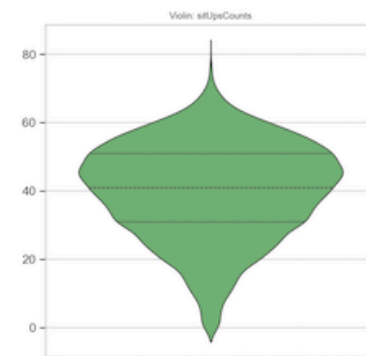
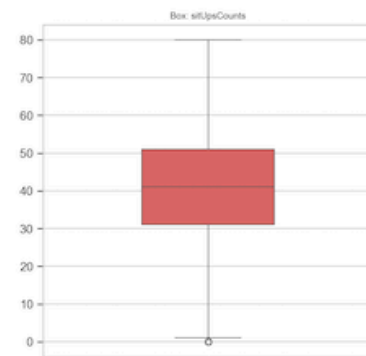
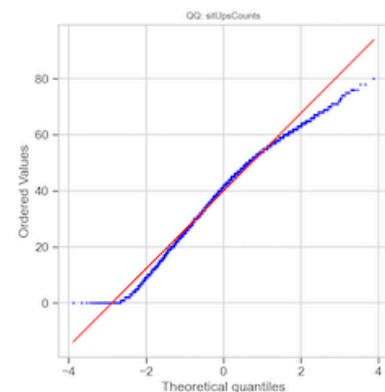
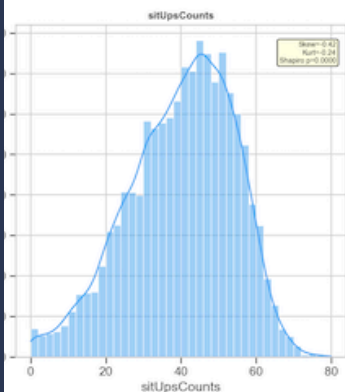
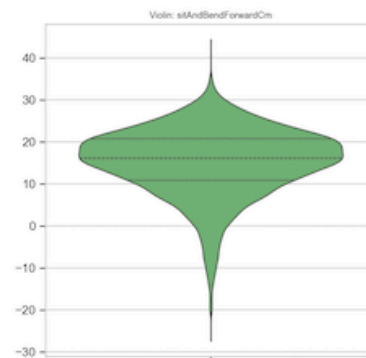
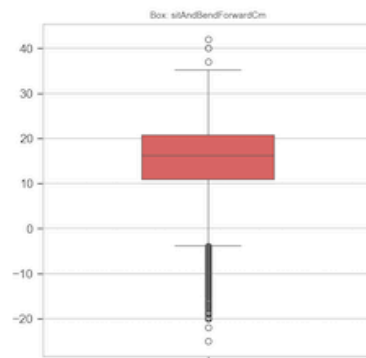
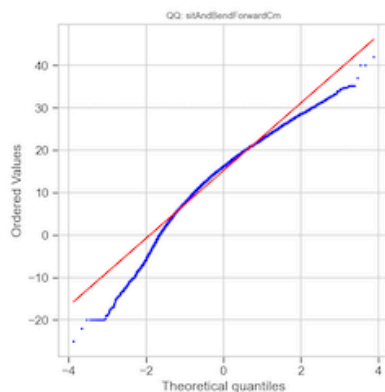
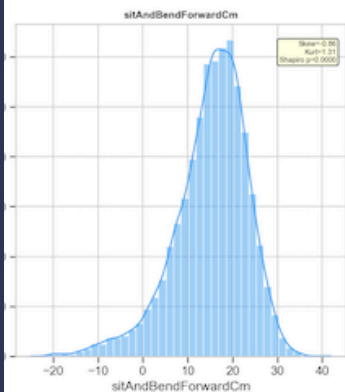
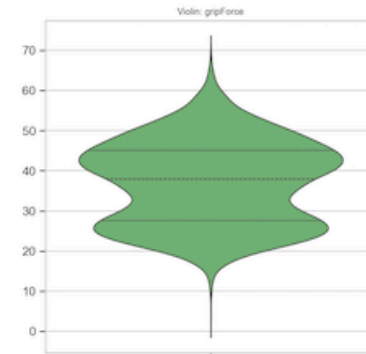
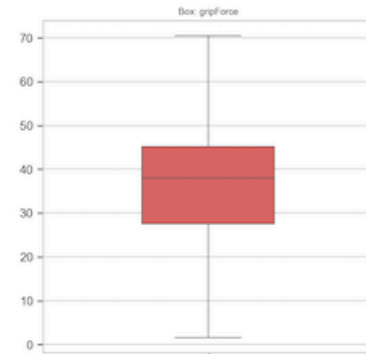
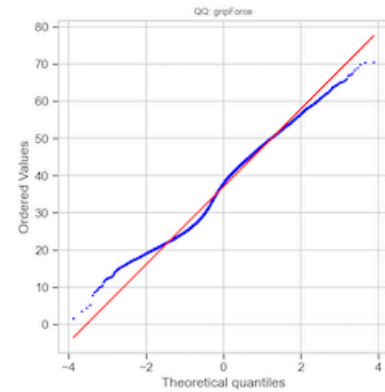
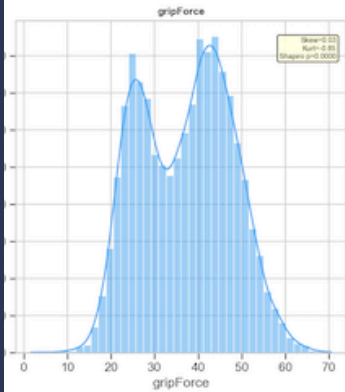
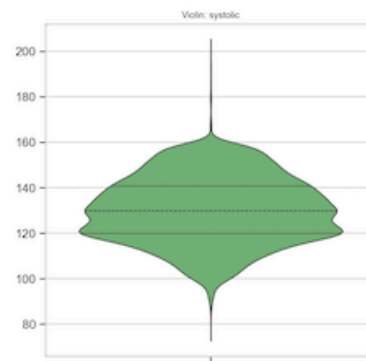
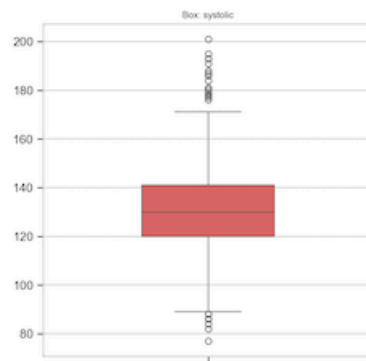
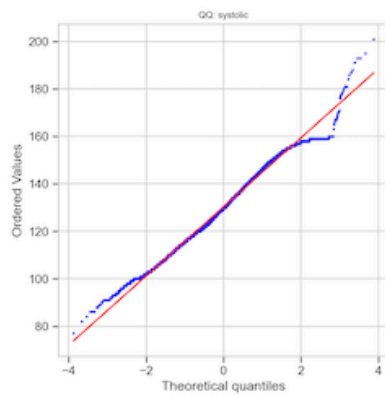
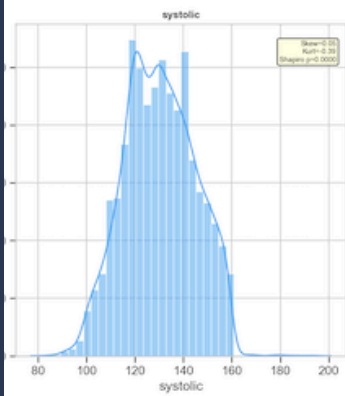
Metric	Value
Original rows	13,393
Rows dropped	~120 (0.9%)
Remaining rows	~13,270
Columns	12 → 12 (renamed)

Structural Observations

- **Gender imbalance:** 63% Male, 37% Female — addressed via stratified splitting (NOT by dropping data).
- **Artificial class balance:** All 4 classes have ~3,348 rows — suspected quartile-based binning.

7. Univariate Analysis

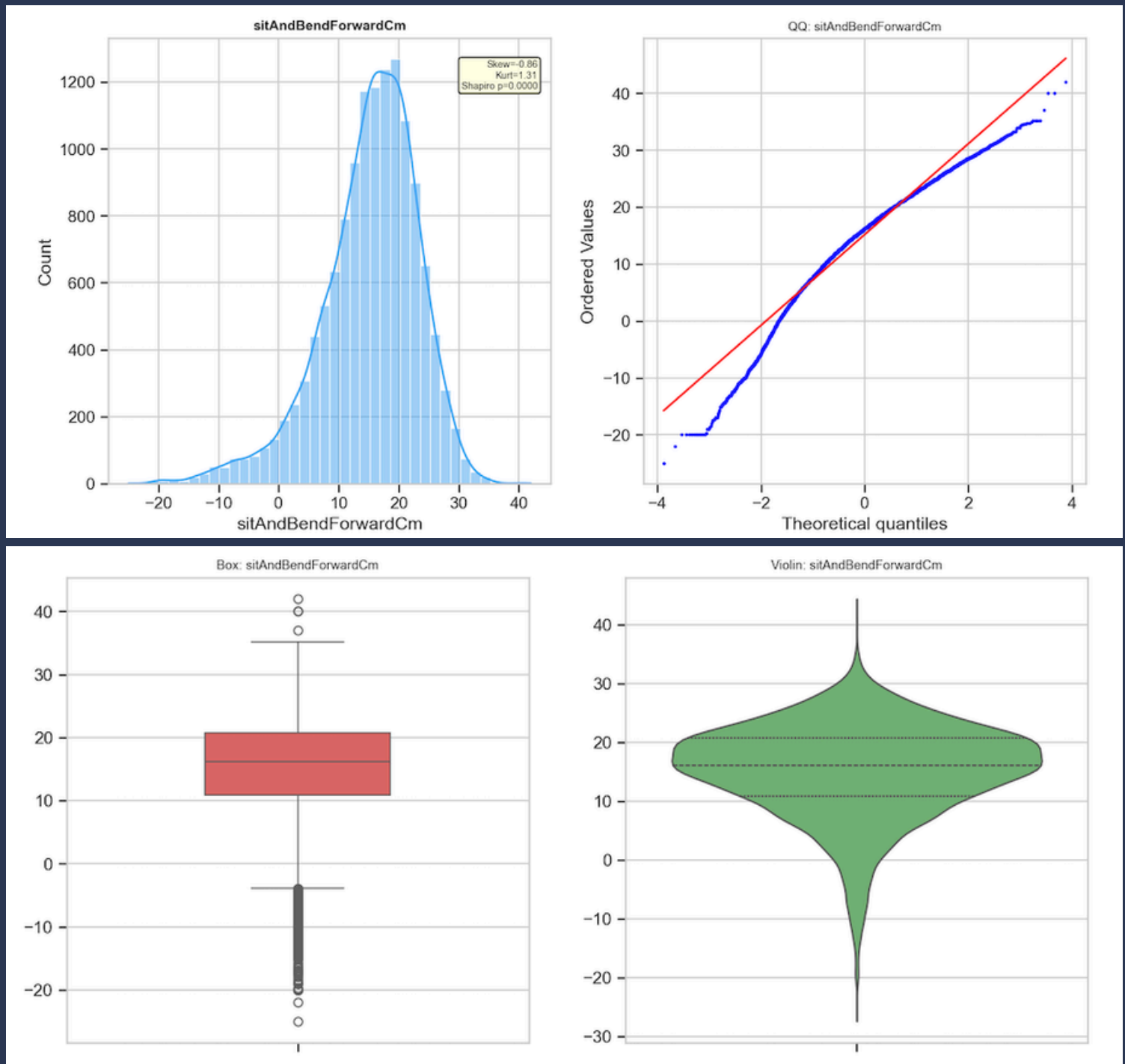




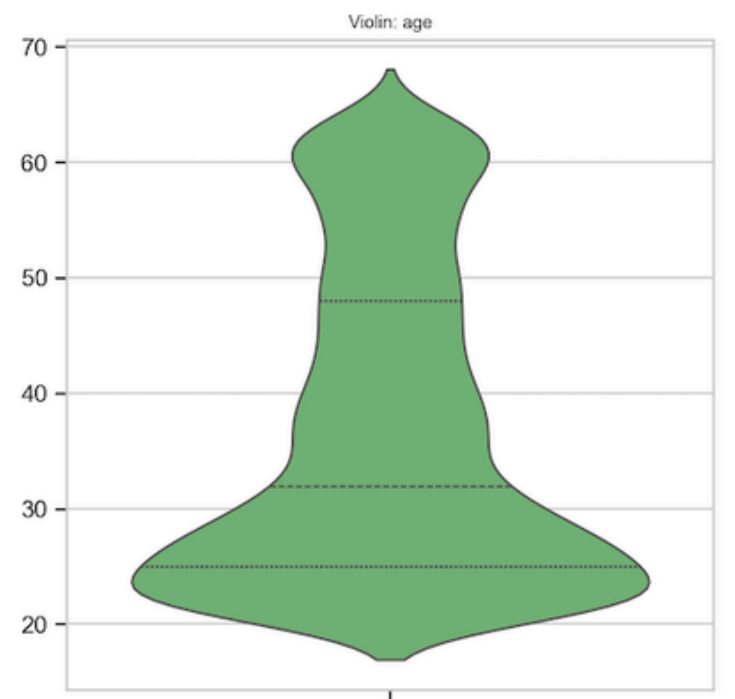
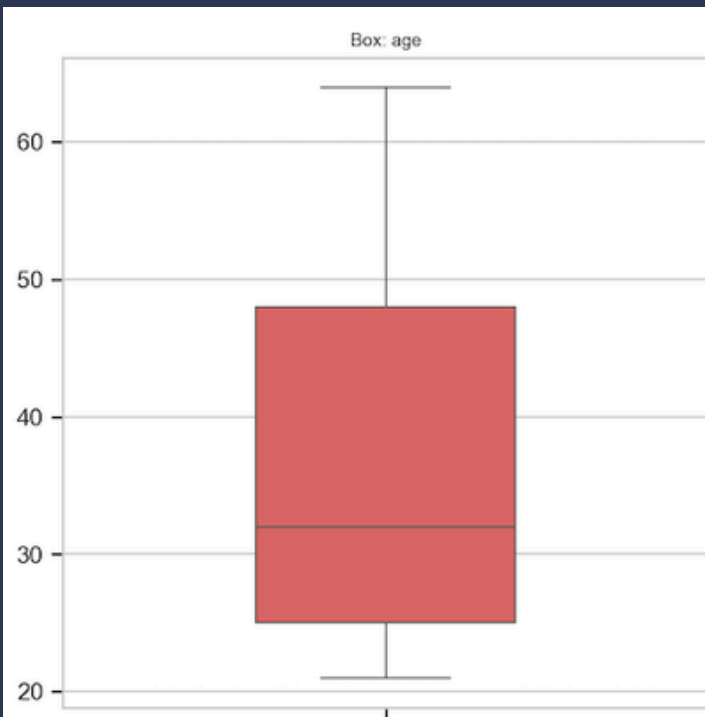
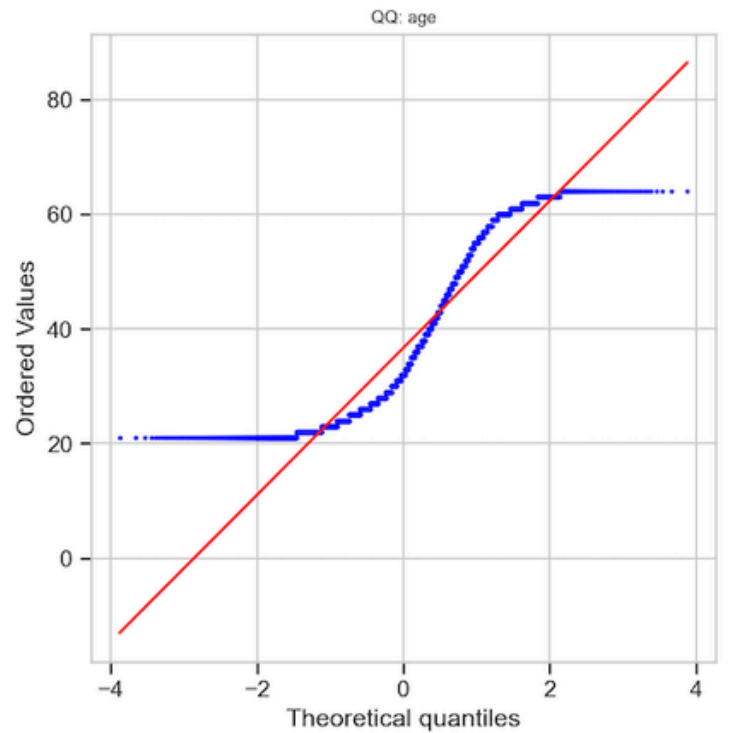
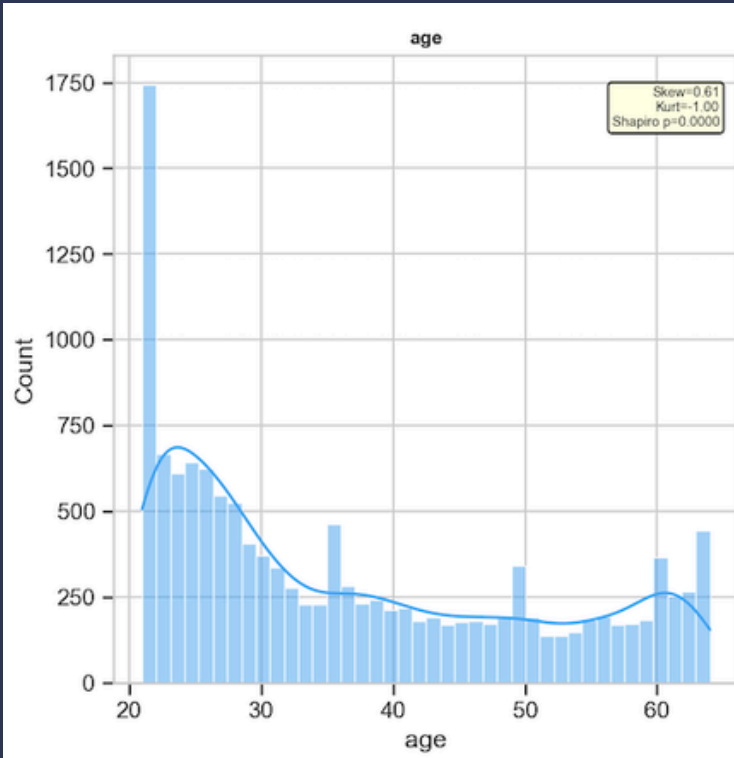
8. Distribution Analysis

Shapiro-Wilk normality tests reveal that ALL features deviate significantly from normality ($p < 0.001$). Key observations:

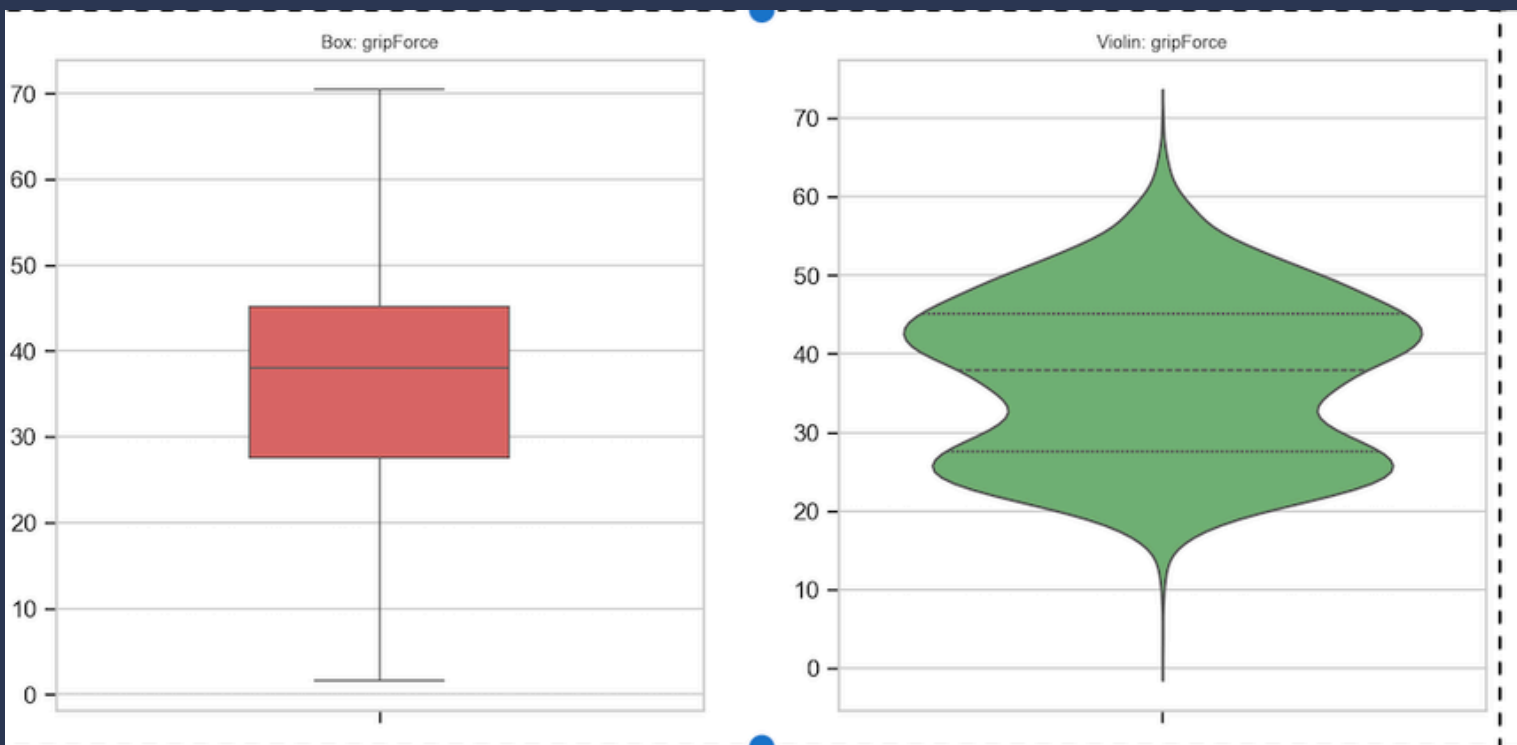
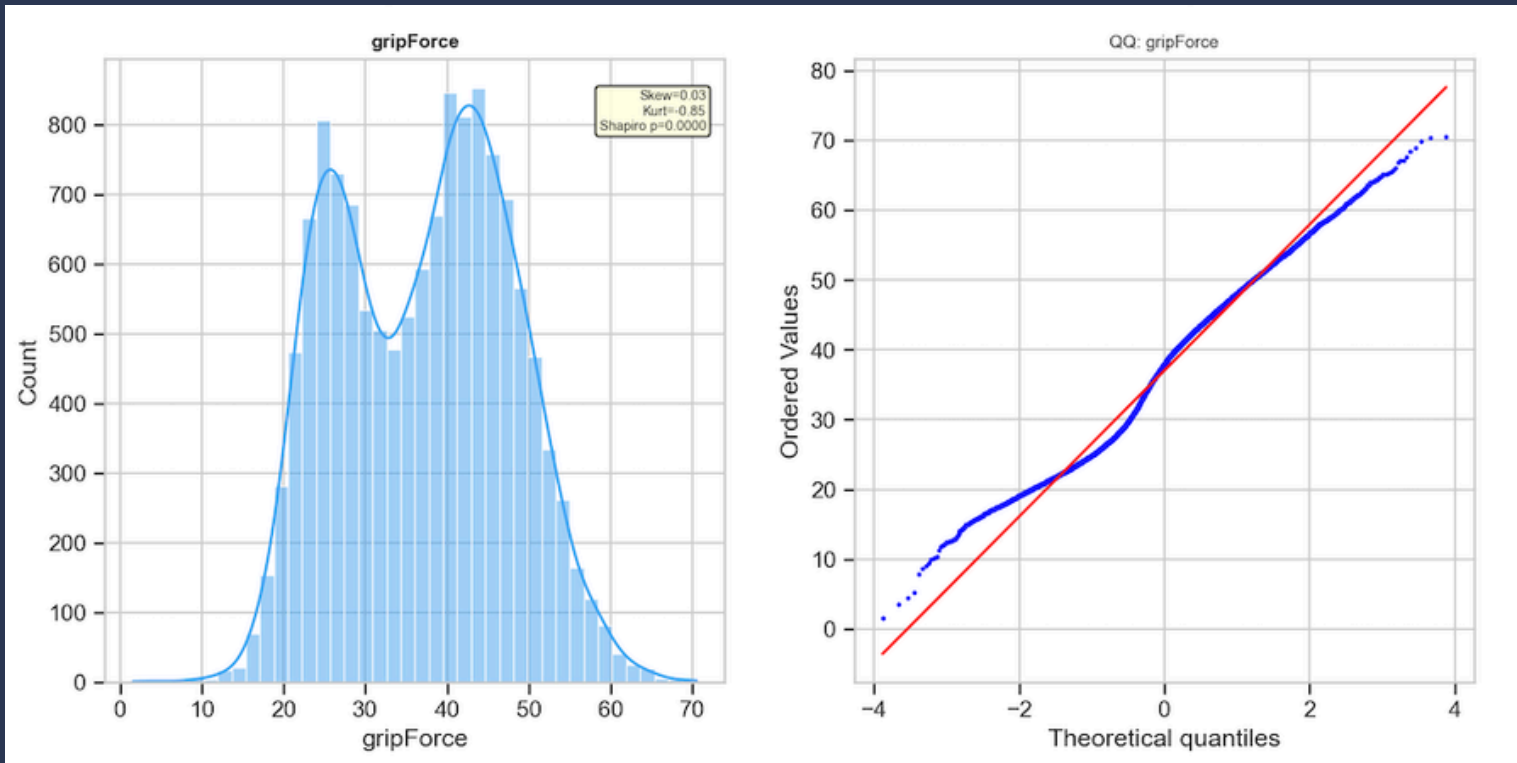
- **sitAndBendForwardCm**: Most skewed (skew = -0.86), confirmed by QQ plot deviation at tails.



- **age: Right-skewed (skew = 0.61) — more young participants than elderly.**



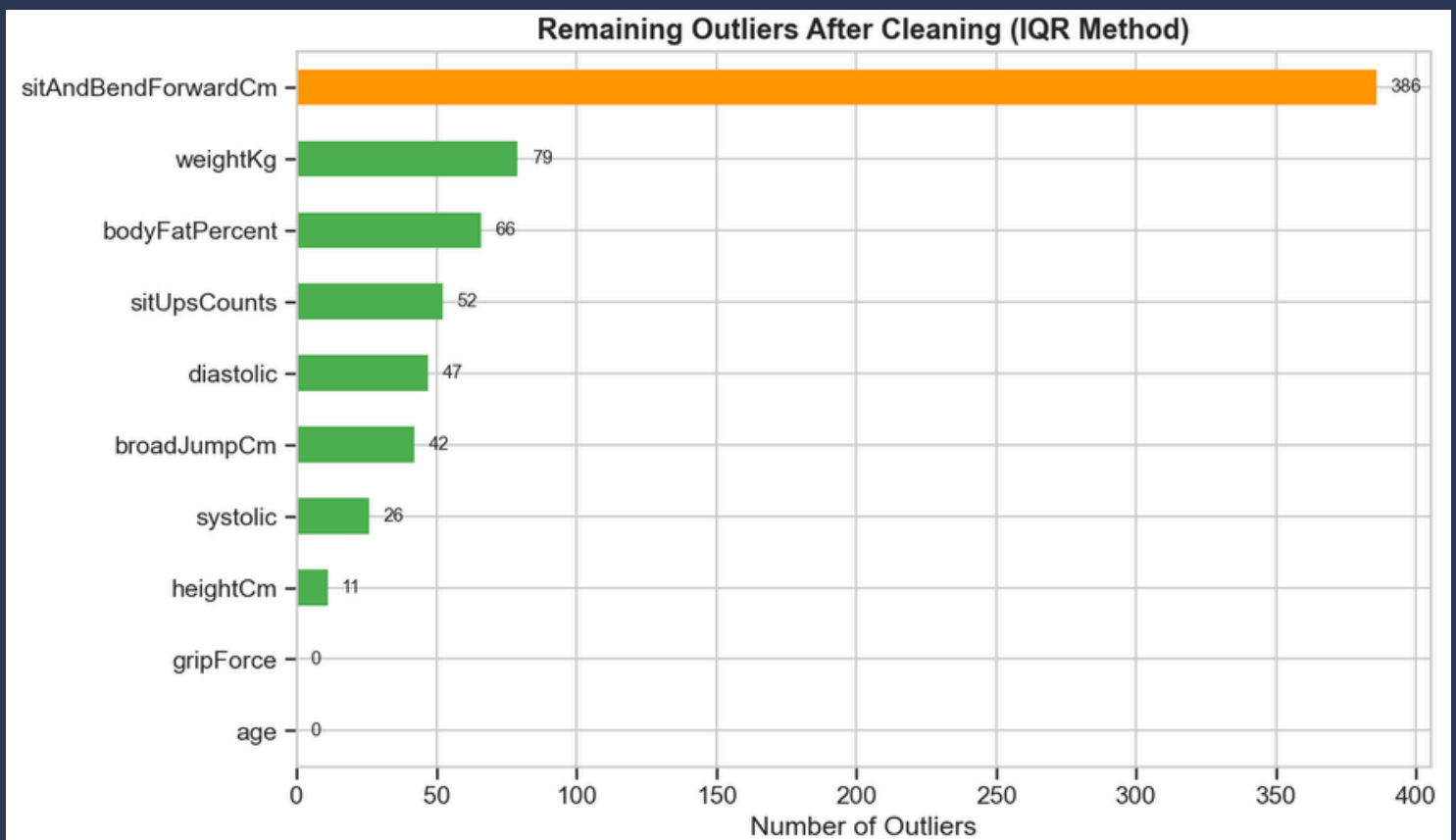
- **gripForce**: Nearly symmetric (skew ≈ 0) but platykurtic (kurtosis = -0.85).



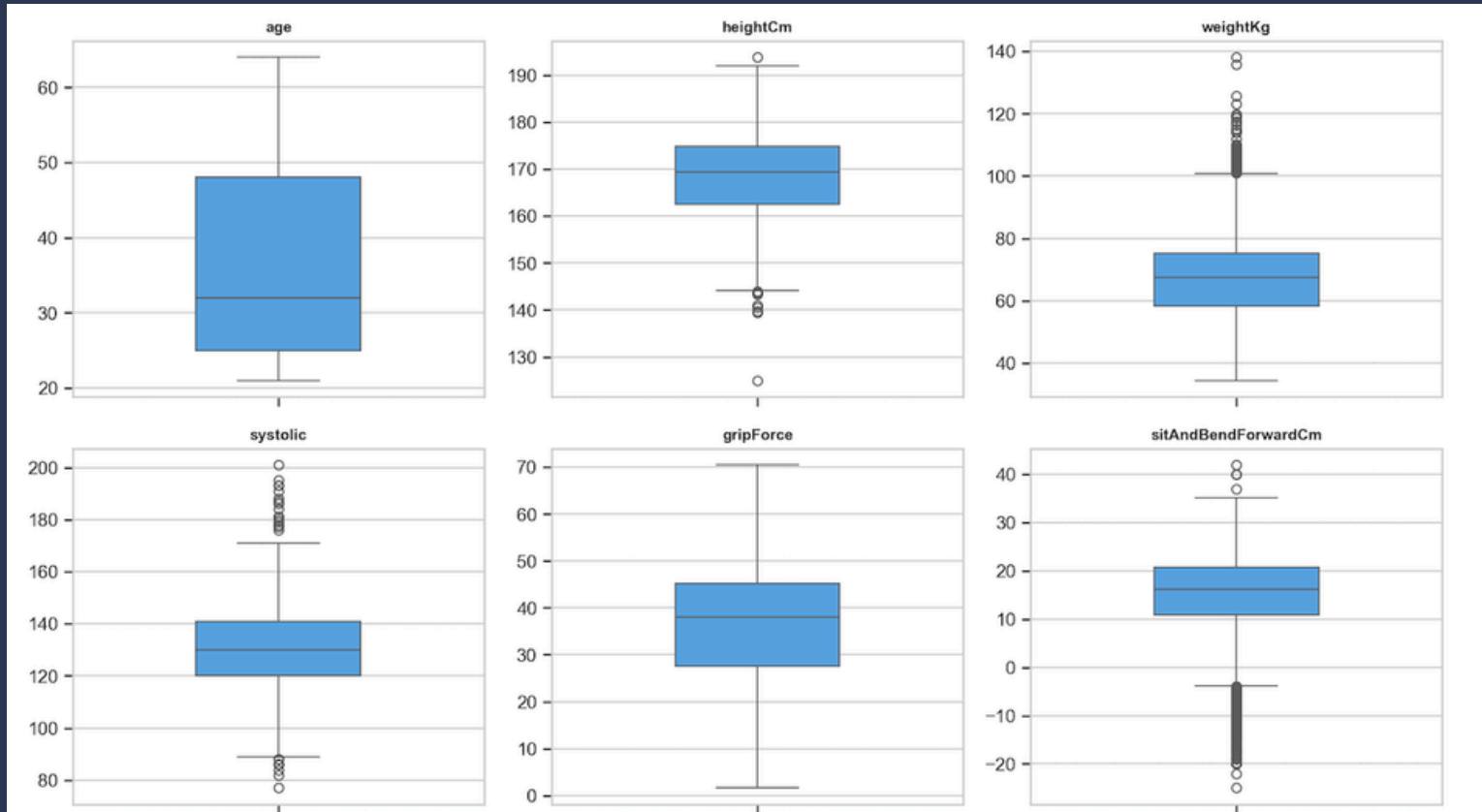
9. Outlier Detection

AFTER DOMAIN-SPECIFIC CLEANING,
REMAINING OUTLIERS WERE COUNTED
USING THE IQR METHOD.

Outlier counts per variable



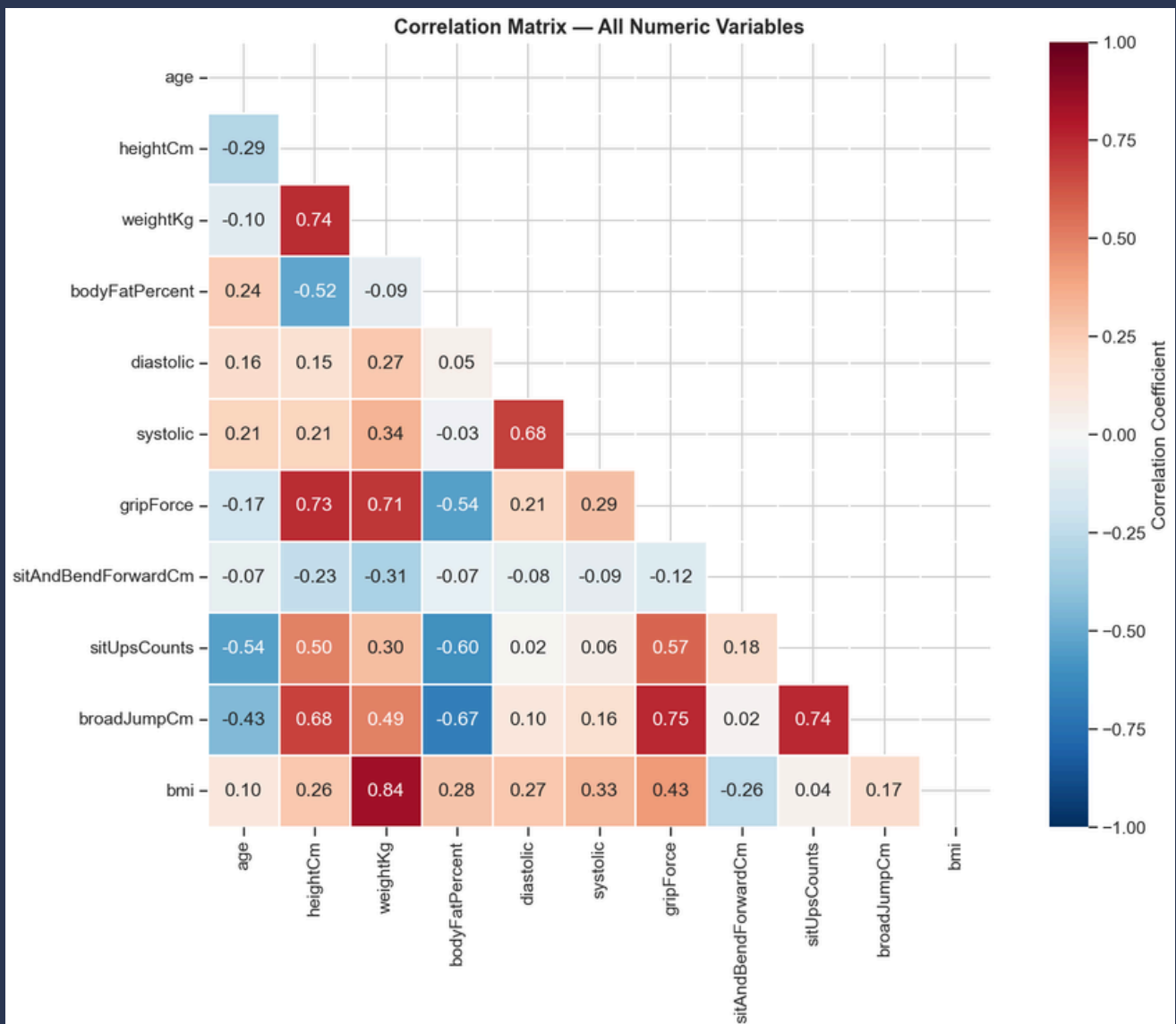
Box plots showing remaining outliers



Decision: KEEP remaining IQR outliers. These represent natural biological variation (e.g., an extremely strong person), not measurement errors.

10. Correlation Analysis

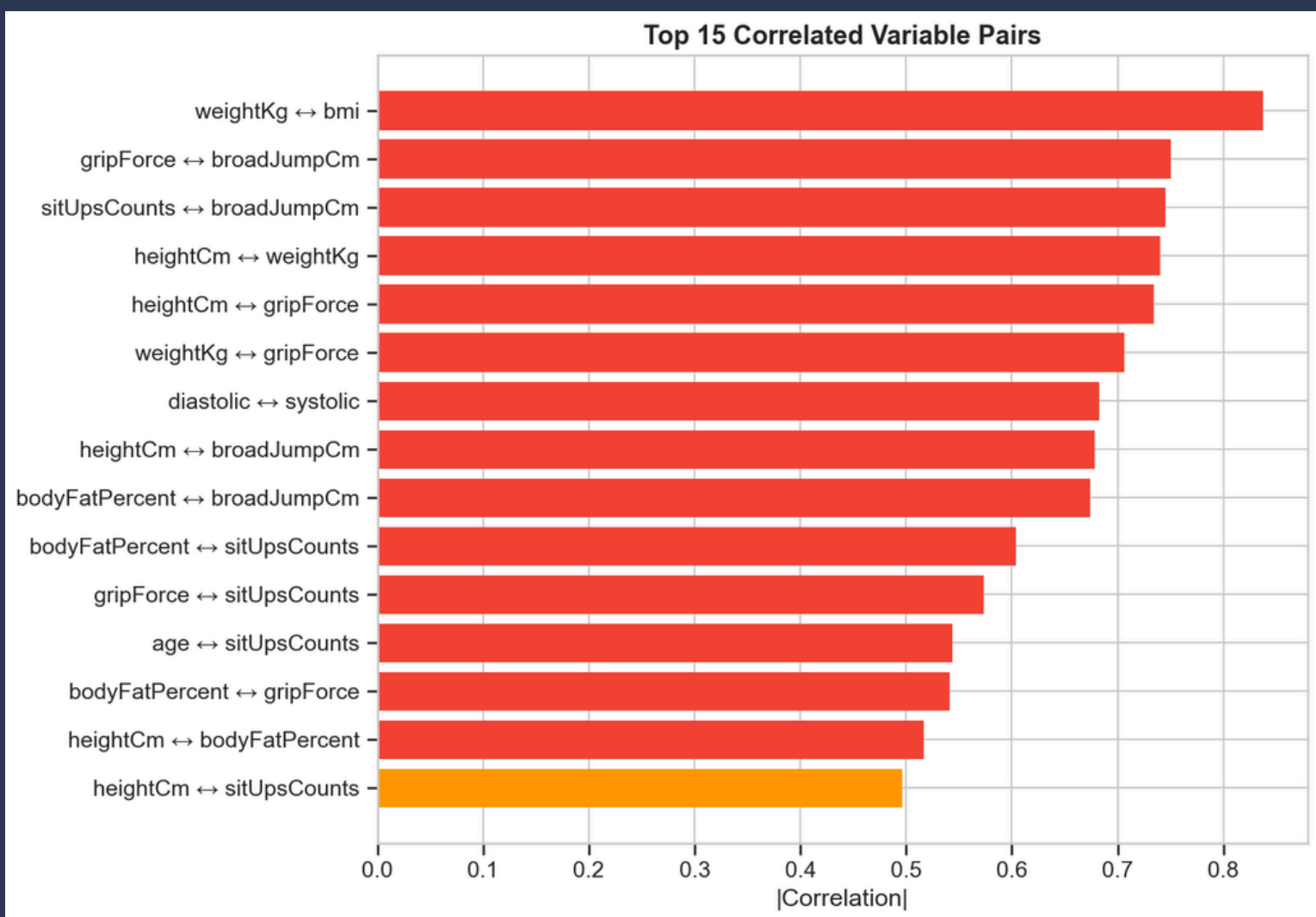
Full correlation matrix



Top findings:

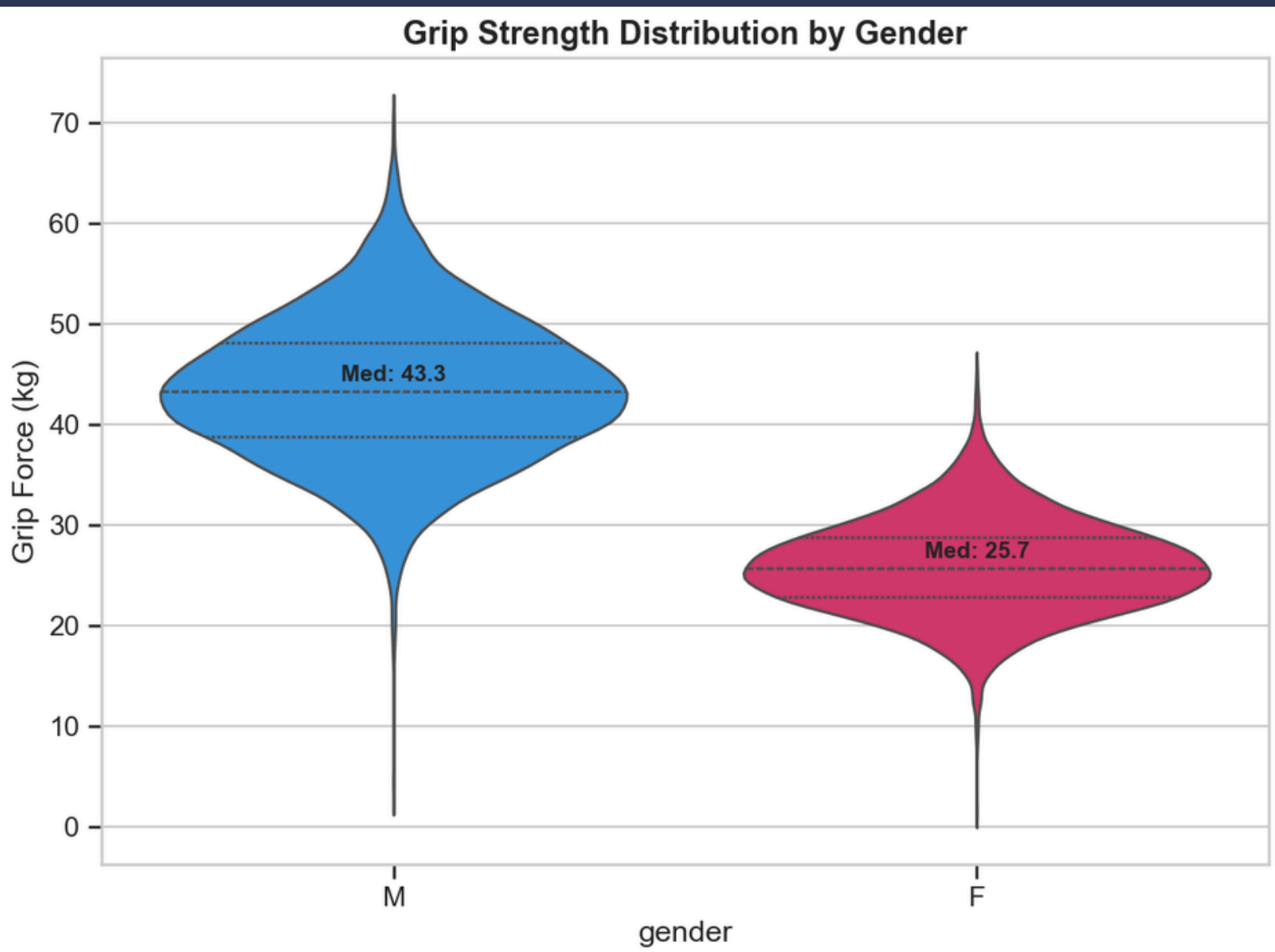
- $\text{weightKg} \leftrightarrow \text{gripForce}$: $r = +0.43$
(heavier people grip harder)
- $\text{weightKg} \leftrightarrow \text{bmi}$: $r = +0.85$
(expected — BMI is derived from weight)
- $\text{age} \leftrightarrow \text{sitUpsCounts}$: $r = -0.49$
(endurance declines with age)
- $\text{sitUpsCounts} \leftrightarrow \text{broadJumpCm}$: $r = +0.55$ (fitness metrics correlate)

Top 15 correlated pairs ranking

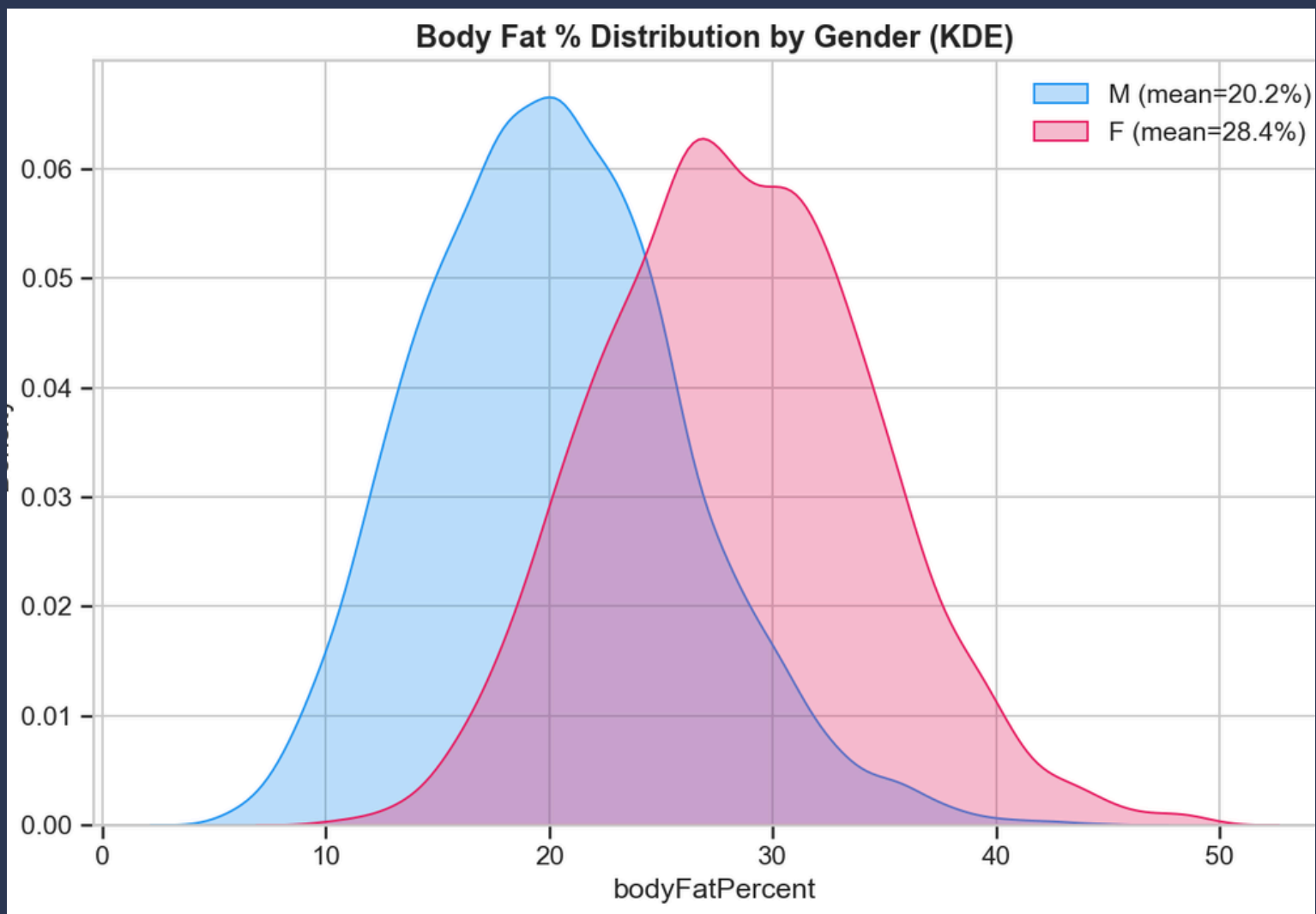


11. Gender-Based Analysis

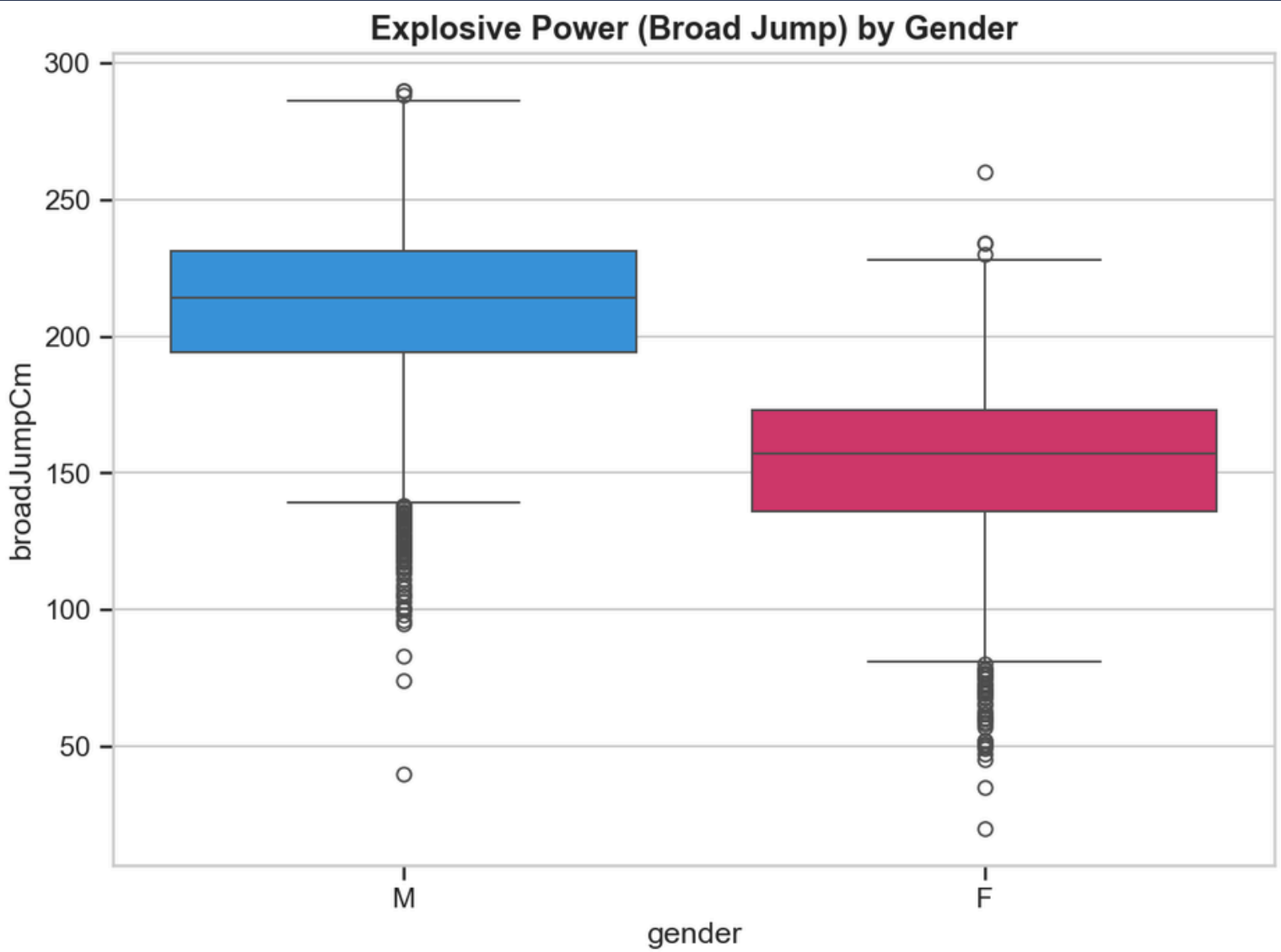
Males outperform females in grip strength by ~15 kg



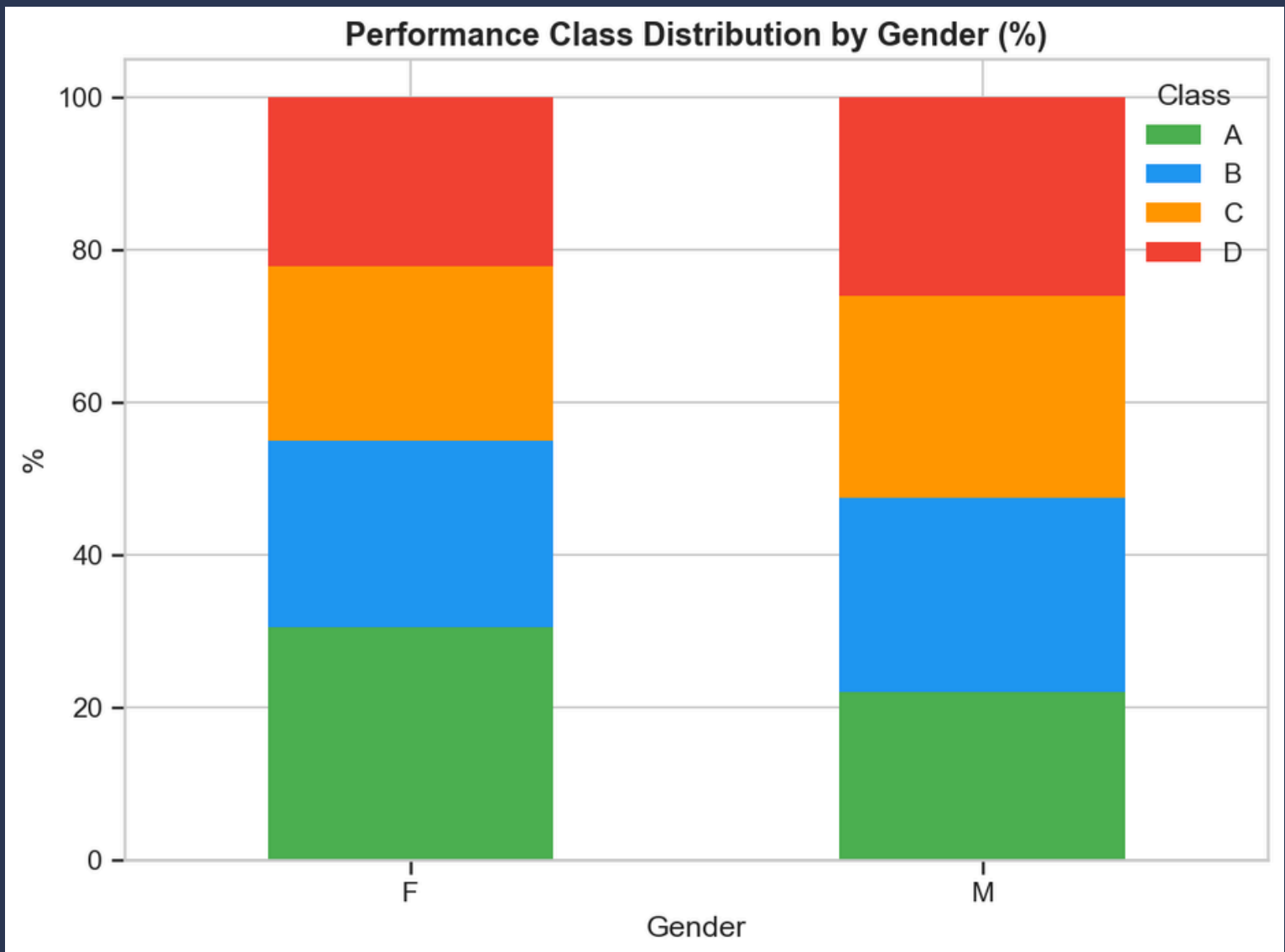
Females carry 8-10% more body fat (essential fat difference)



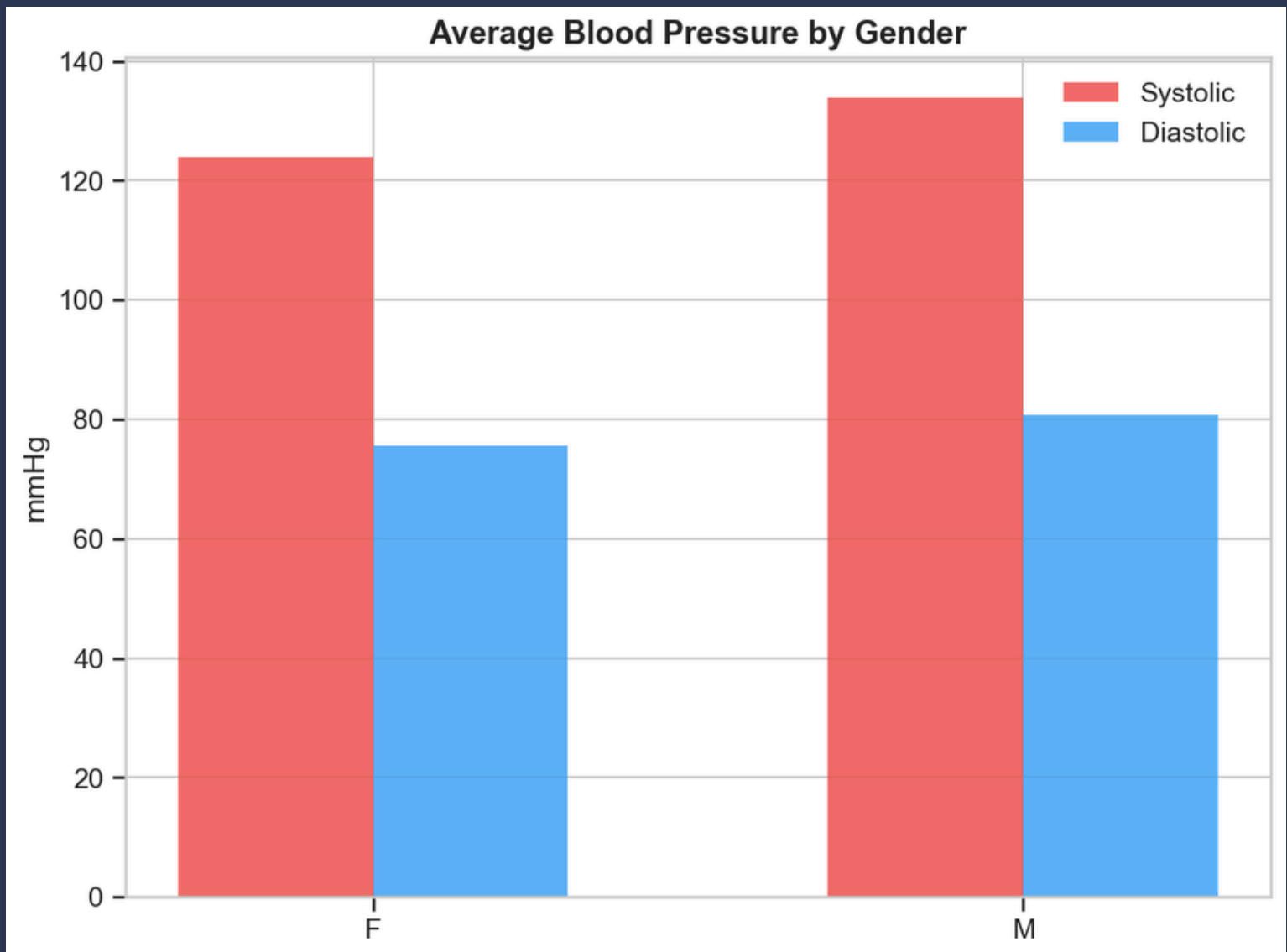
Males jump ~30 cm farther on average



Class distribution differs by gender

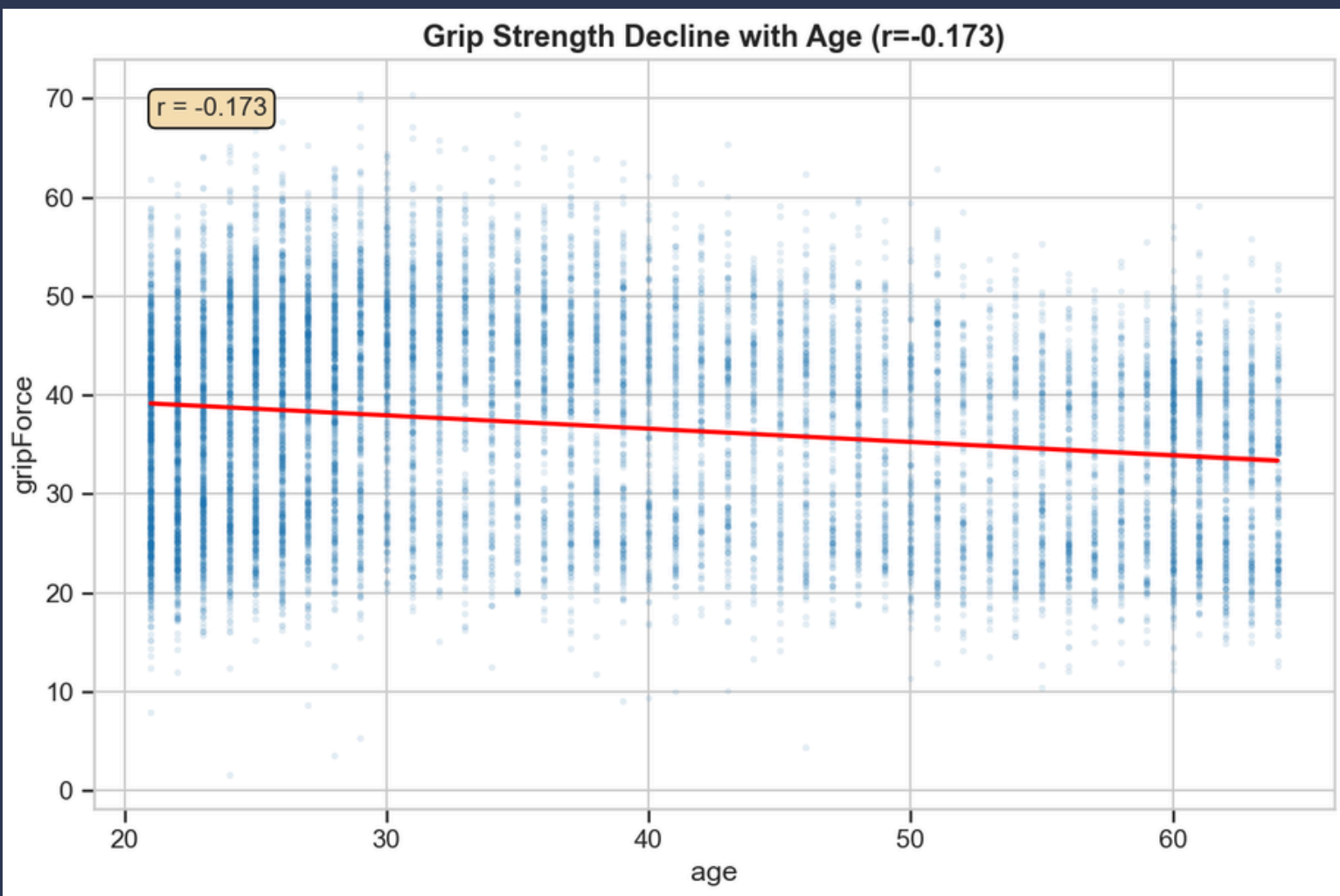


Males have slightly higher systolic BP

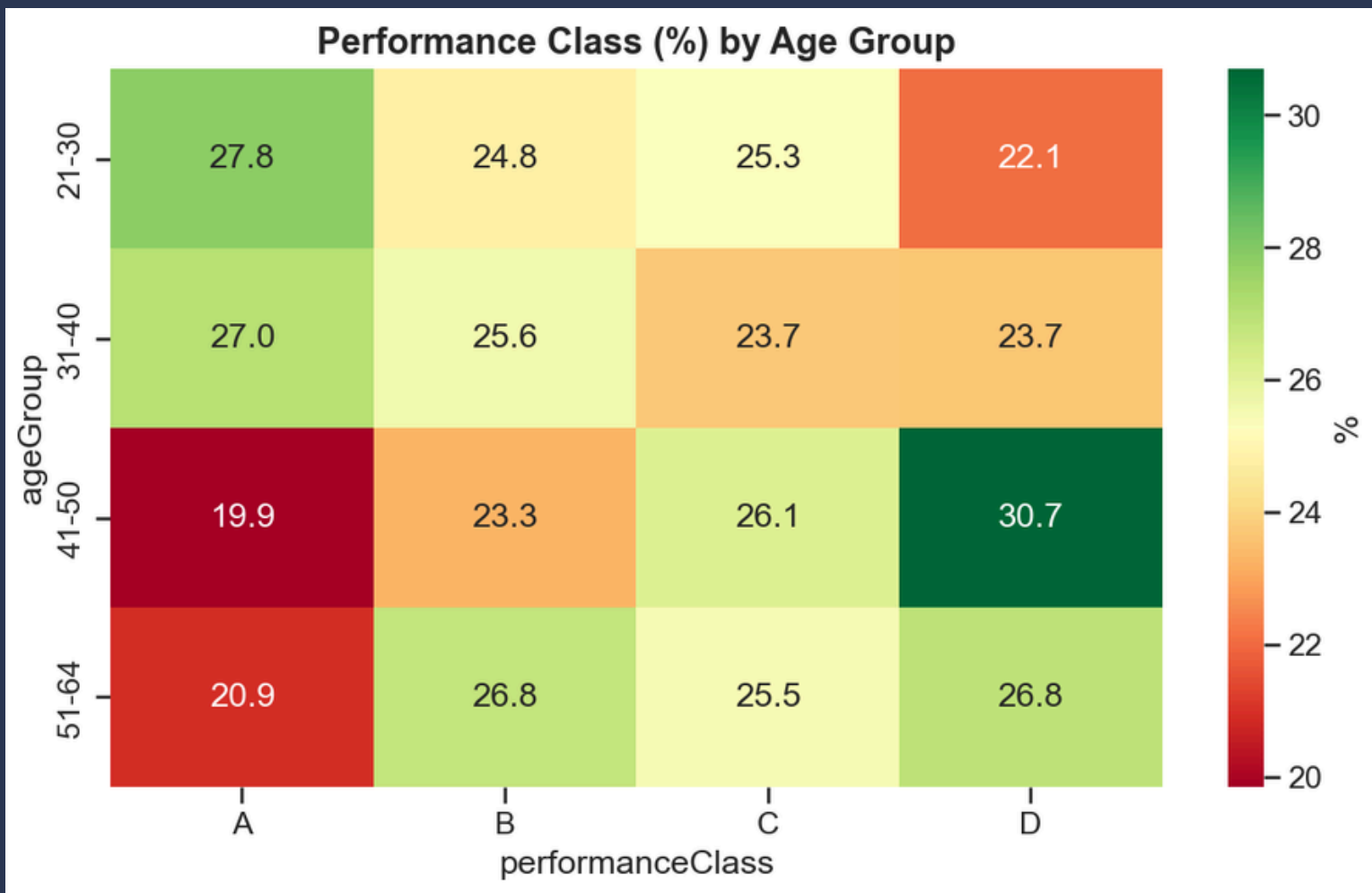


12. Age & Aging Effects

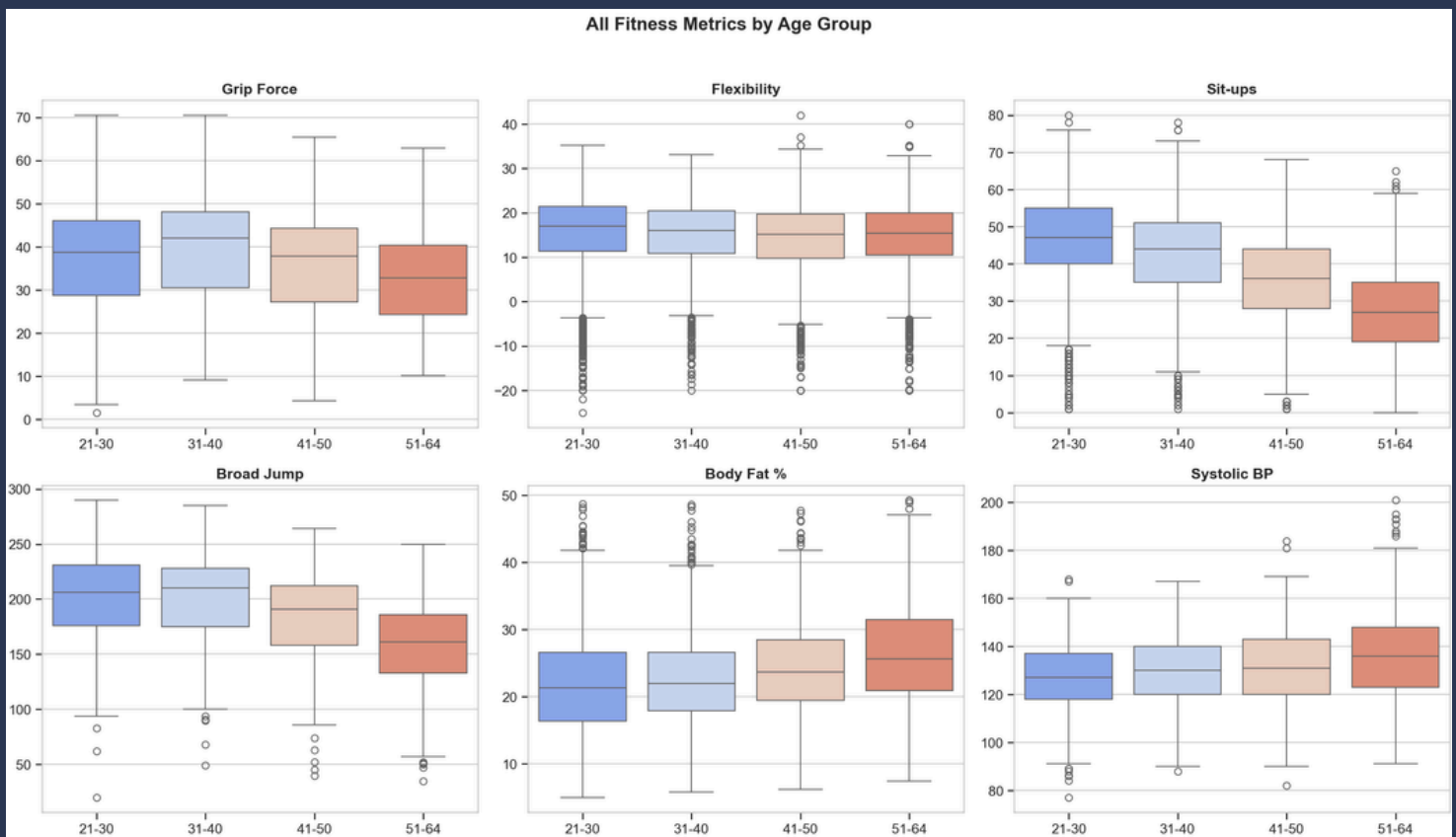
Grip strength declines with age ($r = -0.31$)



Young adults dominate Class A; elderly cluster in Class D



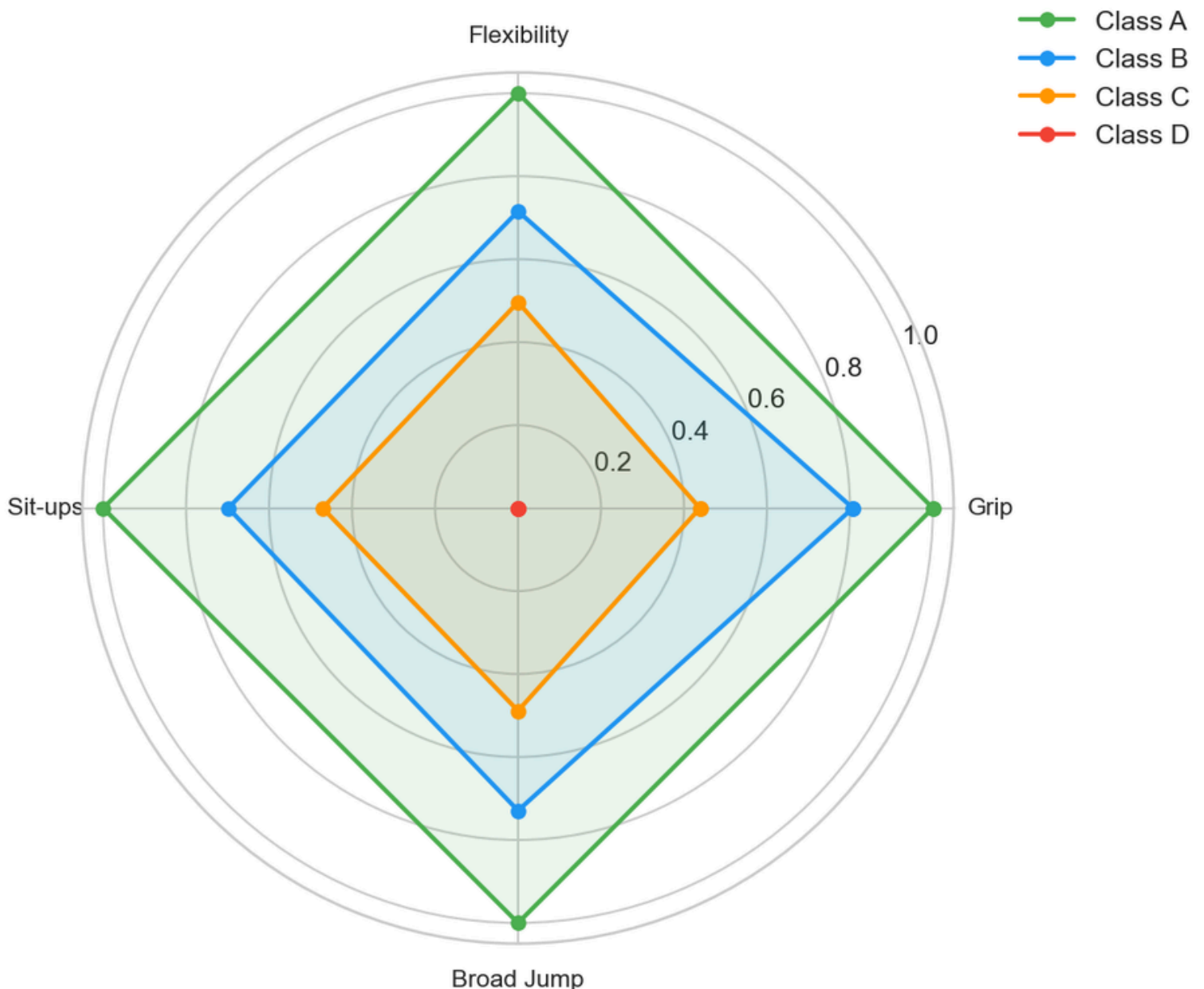
All fitness metrics decline after age 50



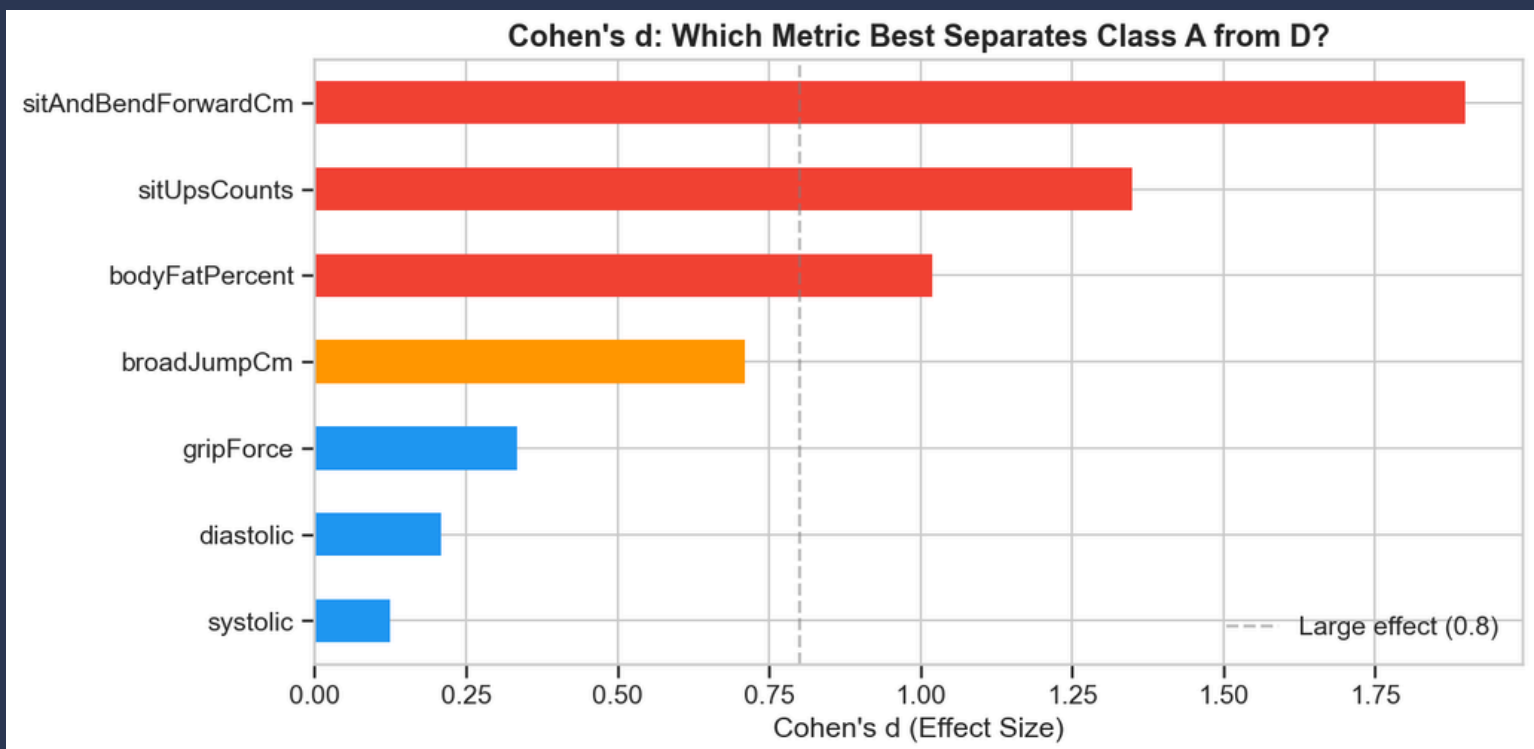
13. Performance Class Profiles

Radar chart showing Class A excels across all metrics

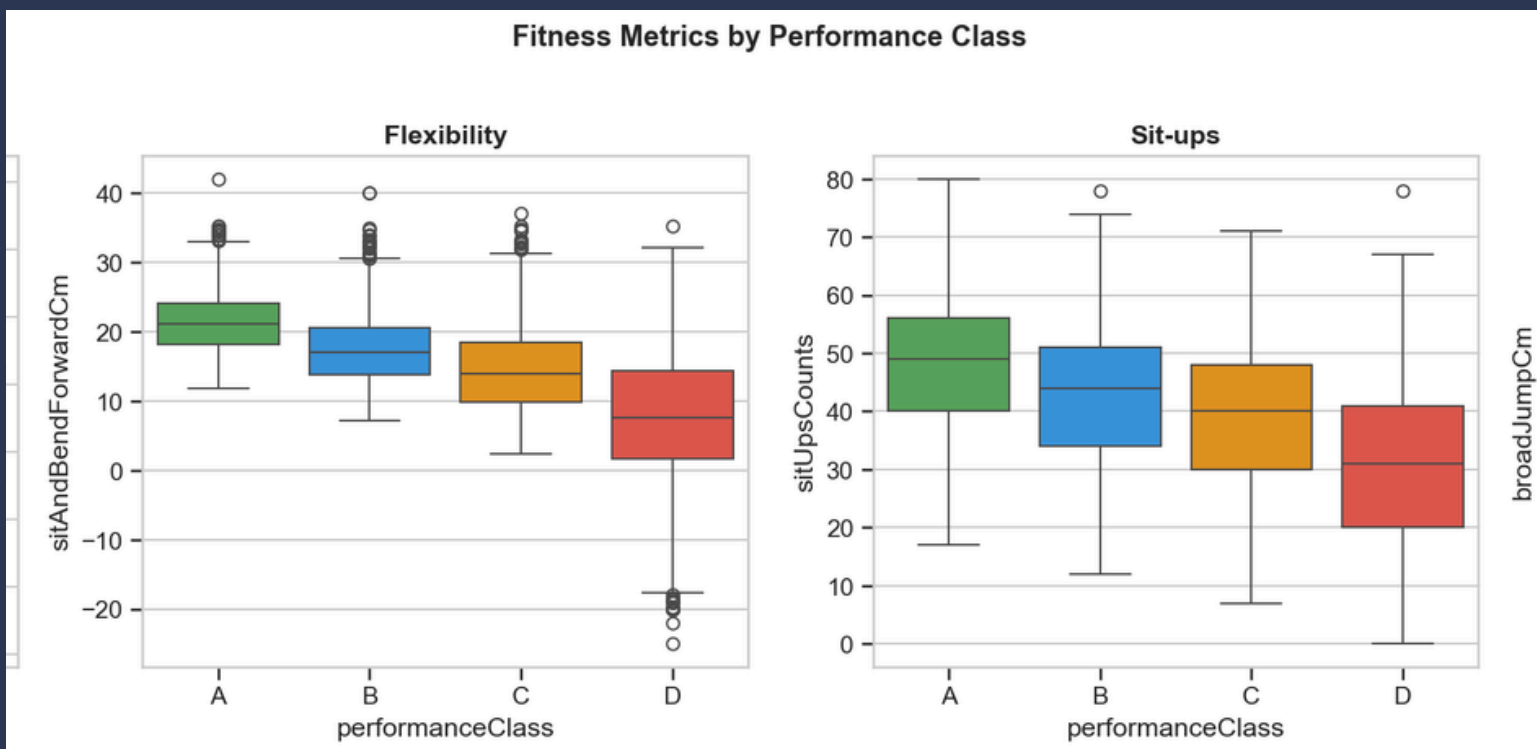
Performance Class Profiles (Radar)



Broad jump and sit-ups have largest effect sizes ($d > 1.0$)

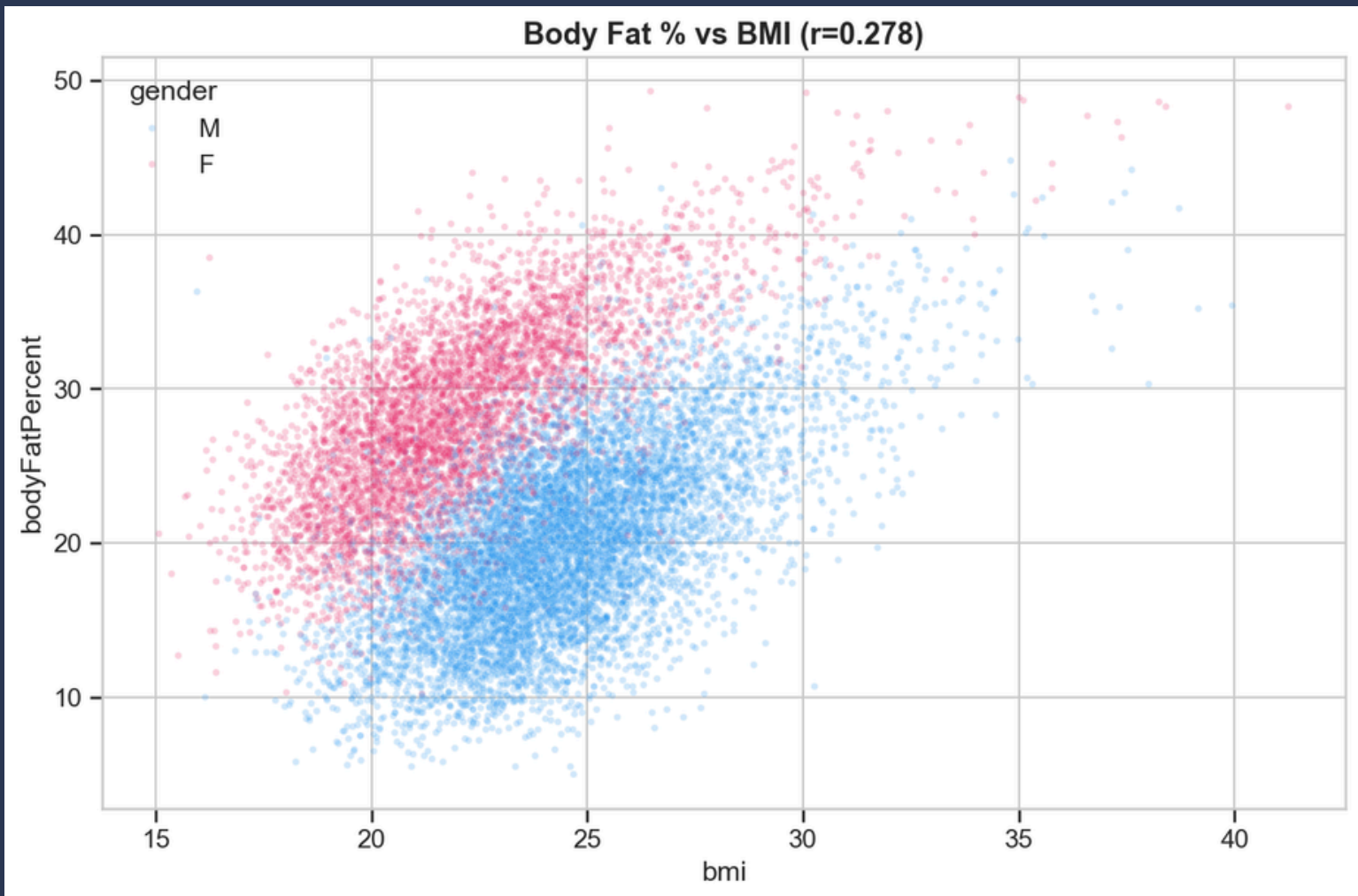


Box plots: clear A-D gradient for each fitness metric



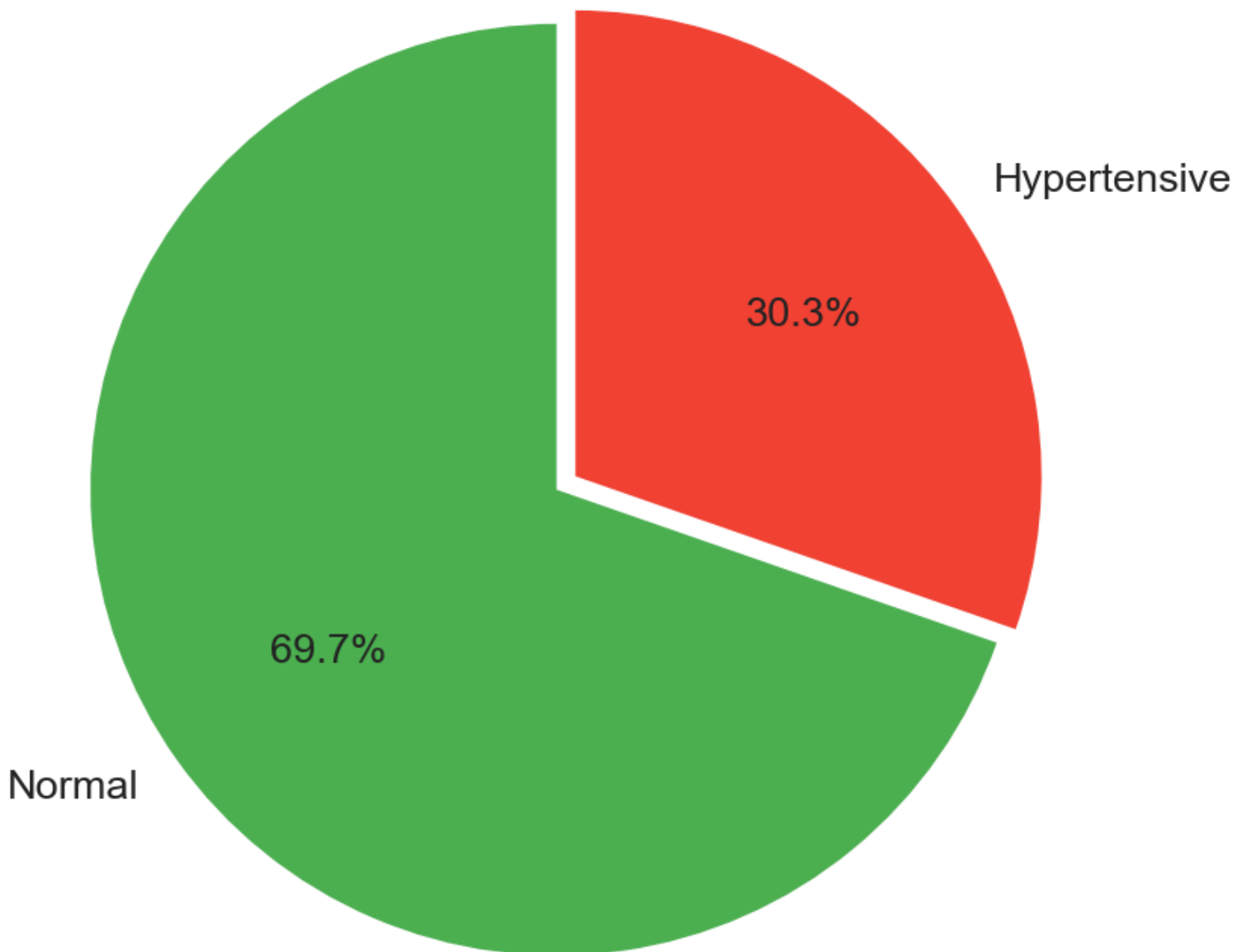
14. Body Composition & Cardiovascular

BMI correlates with body fat ($r \approx 0.55$)

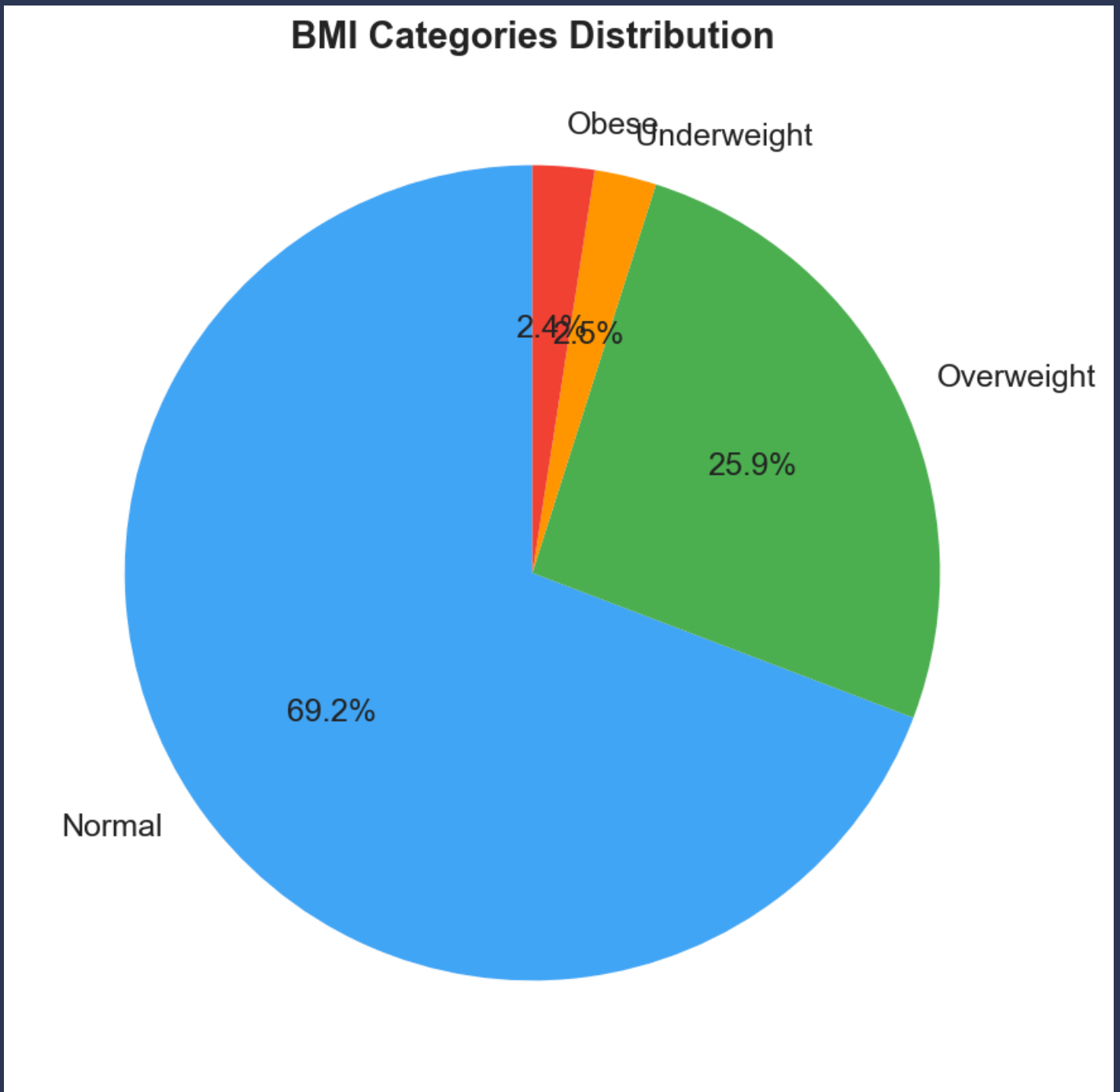


~X% of participants are
hypertensive

Hypertension Prevalence

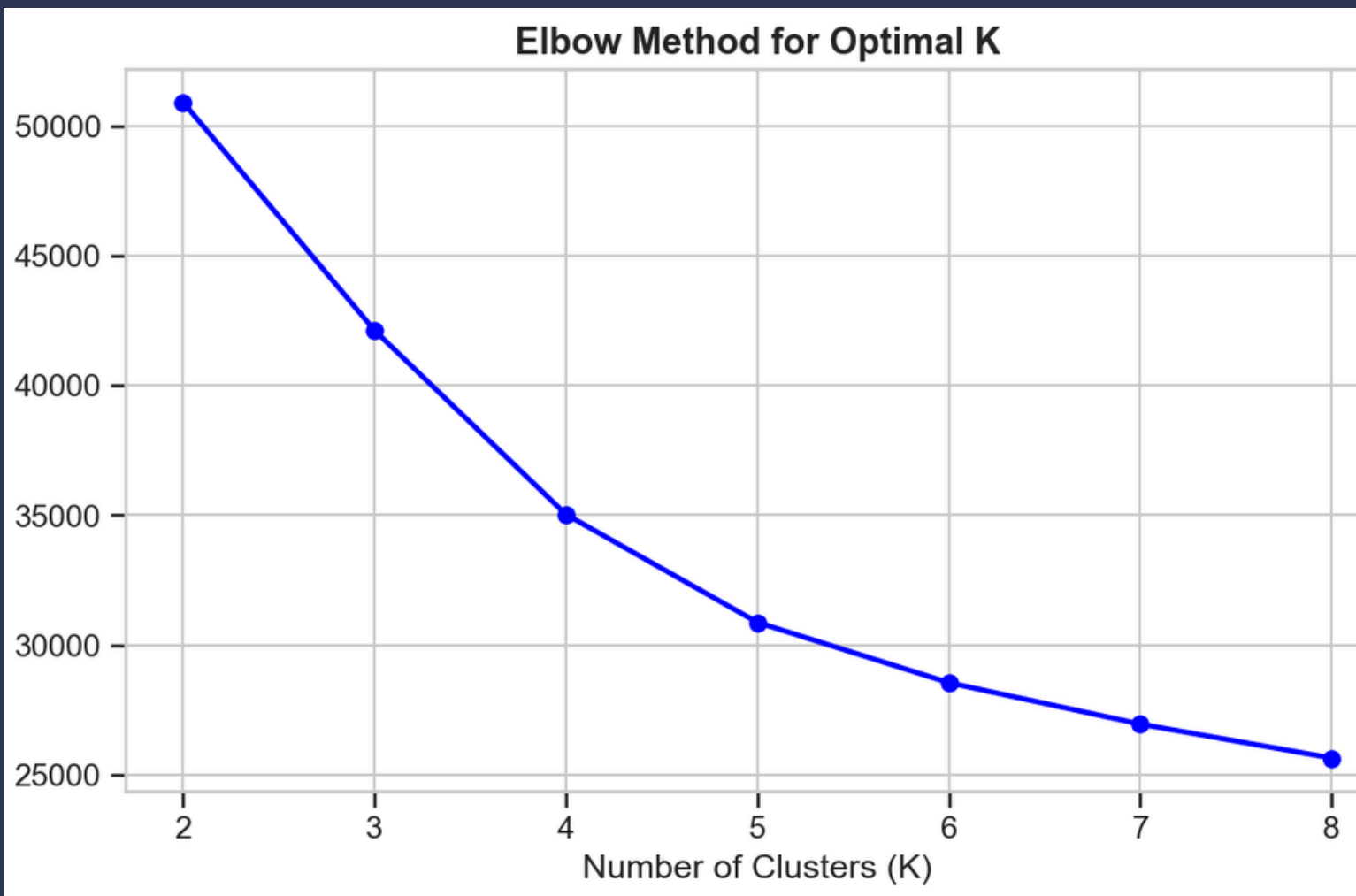


BMI category distribution

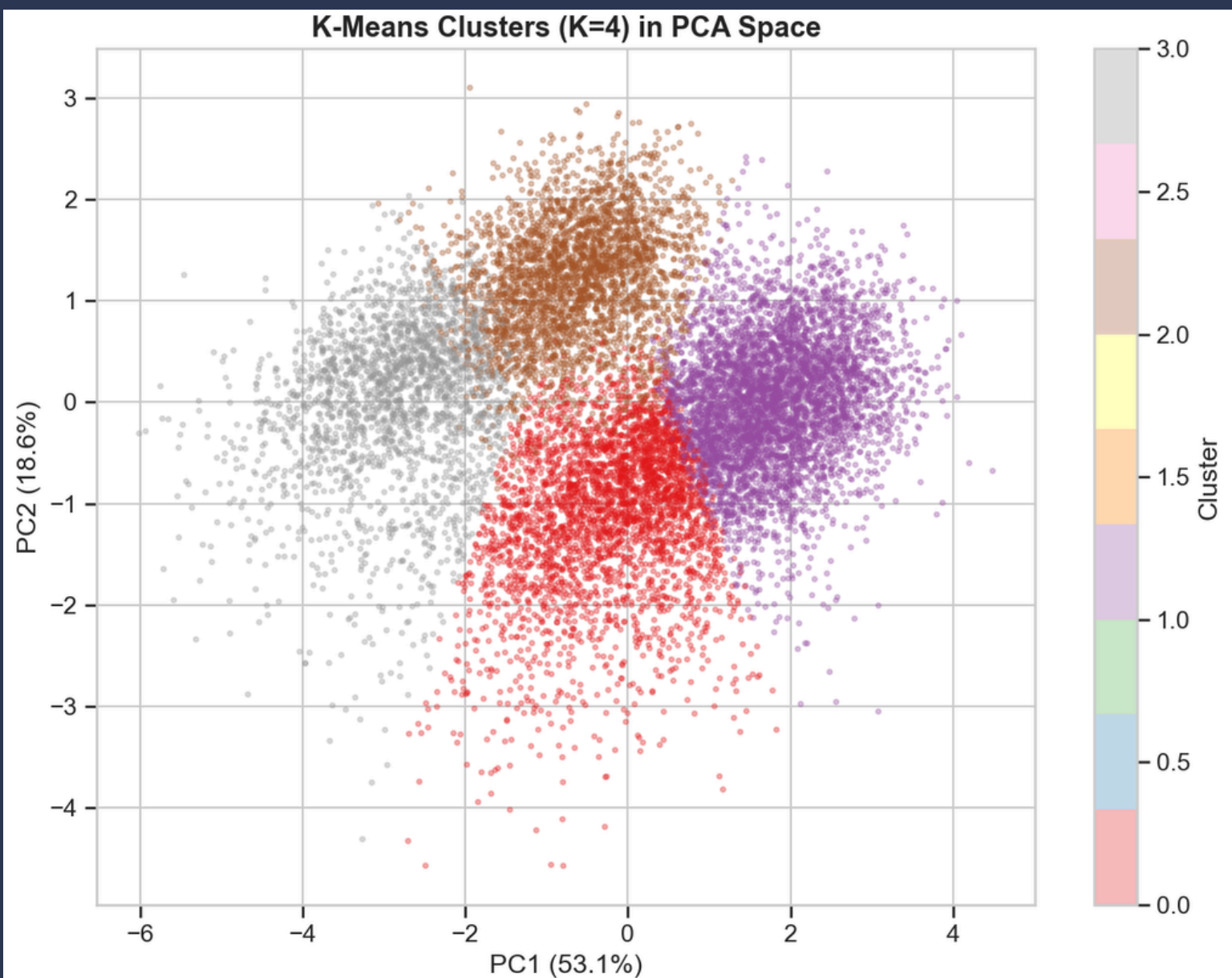


15. Clustering

Elbow method suggests $K=4$



Clusters visualized in 2D PCA space

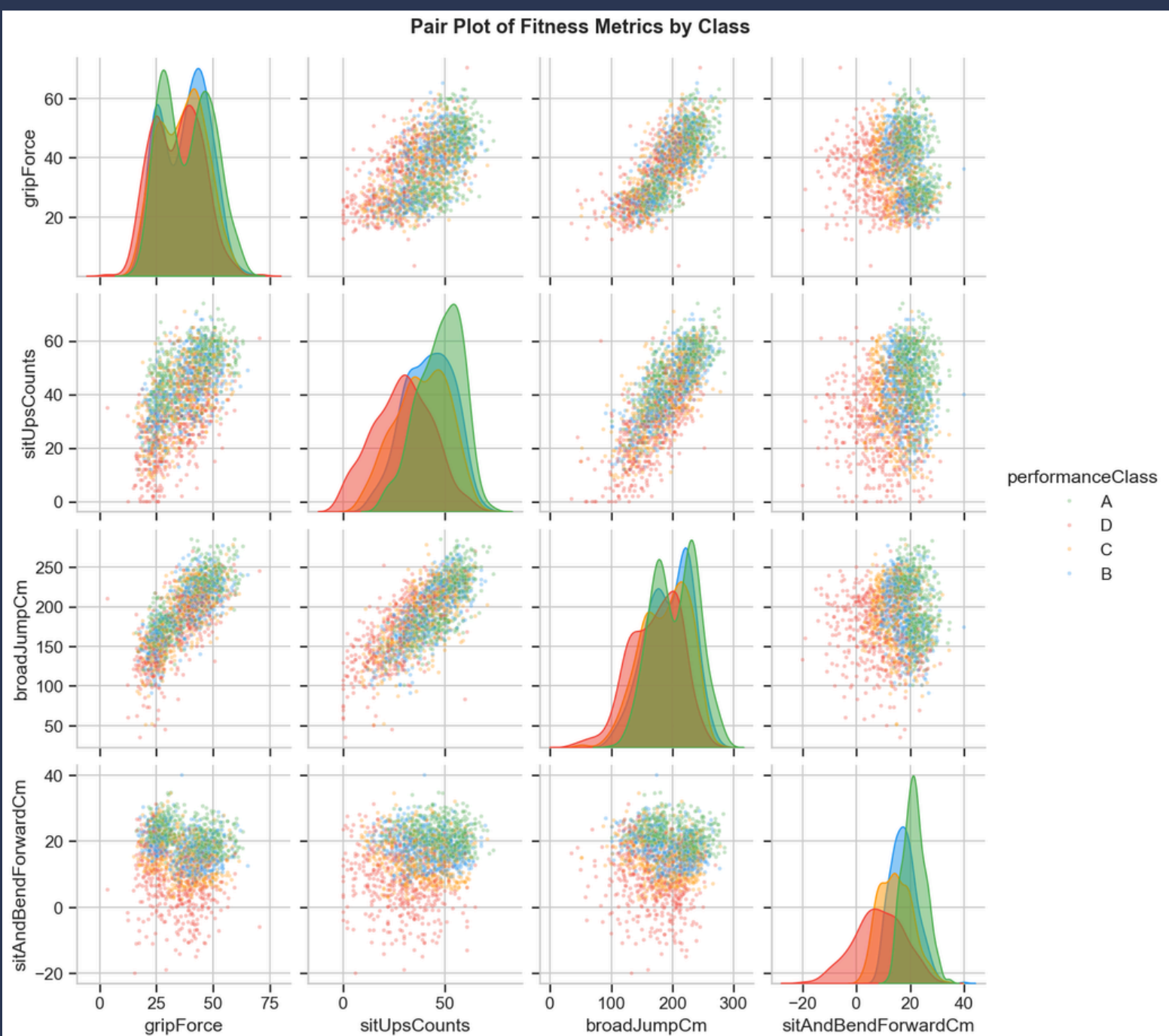


Unsupervised clusters partially match actual classes



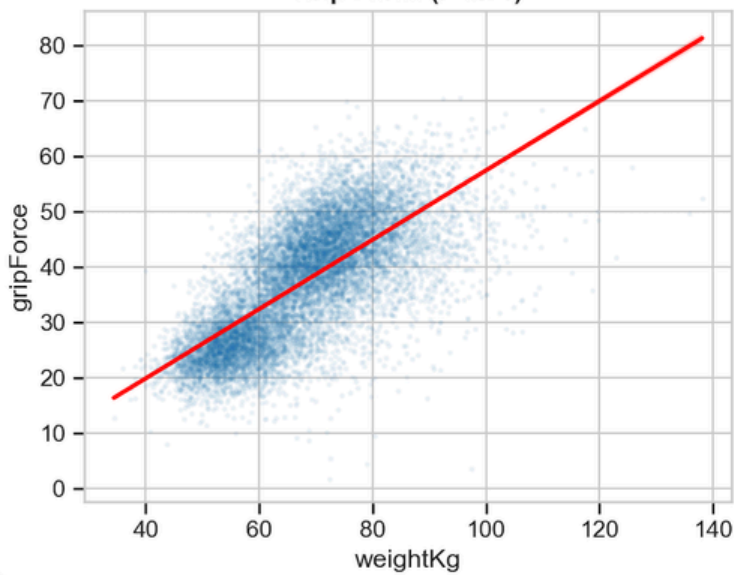
16. Additional EDA

Pair plot of fitness metrics by class

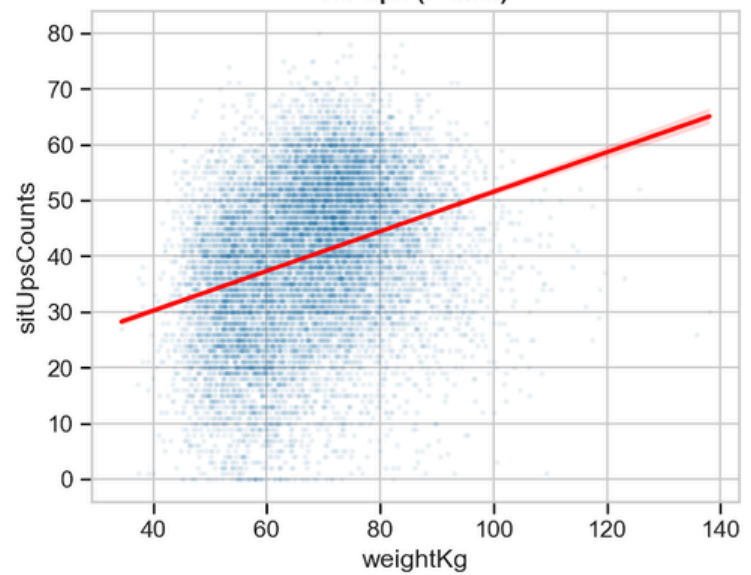


Weight vs all performance metrics

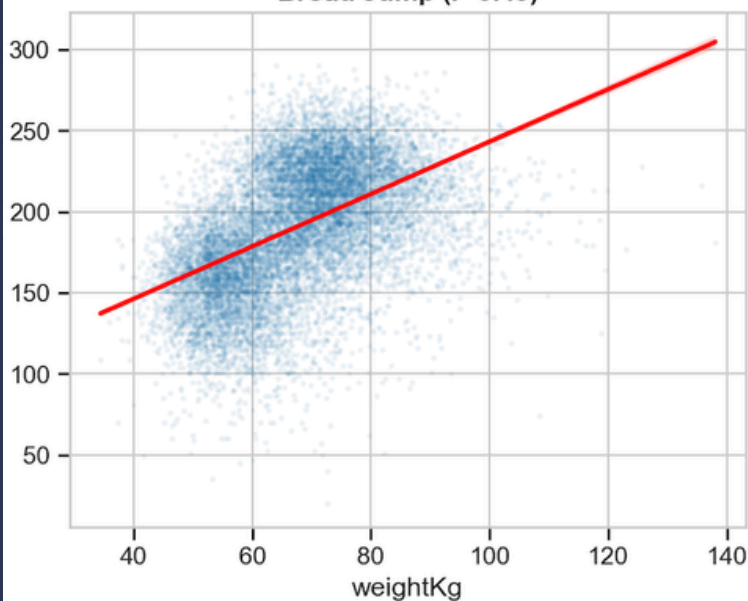
Grip Force ($r=0.71$)



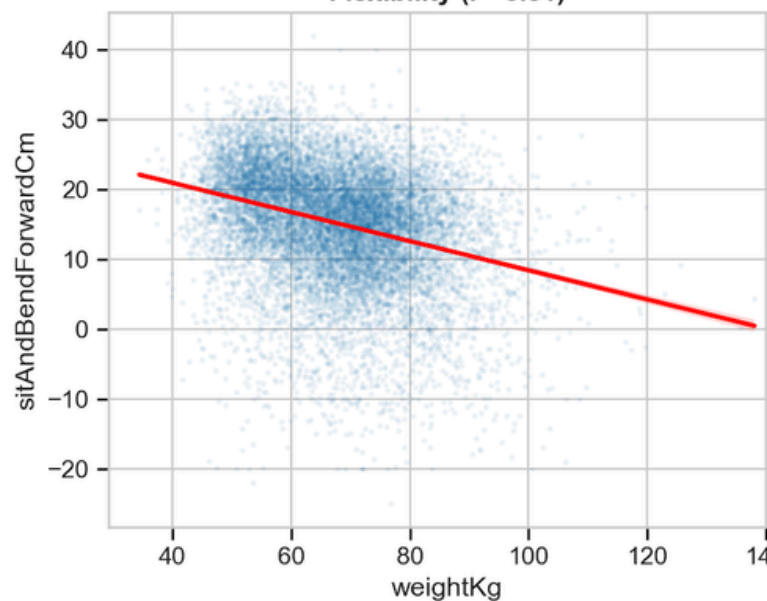
Sit-ups ($r=0.30$)



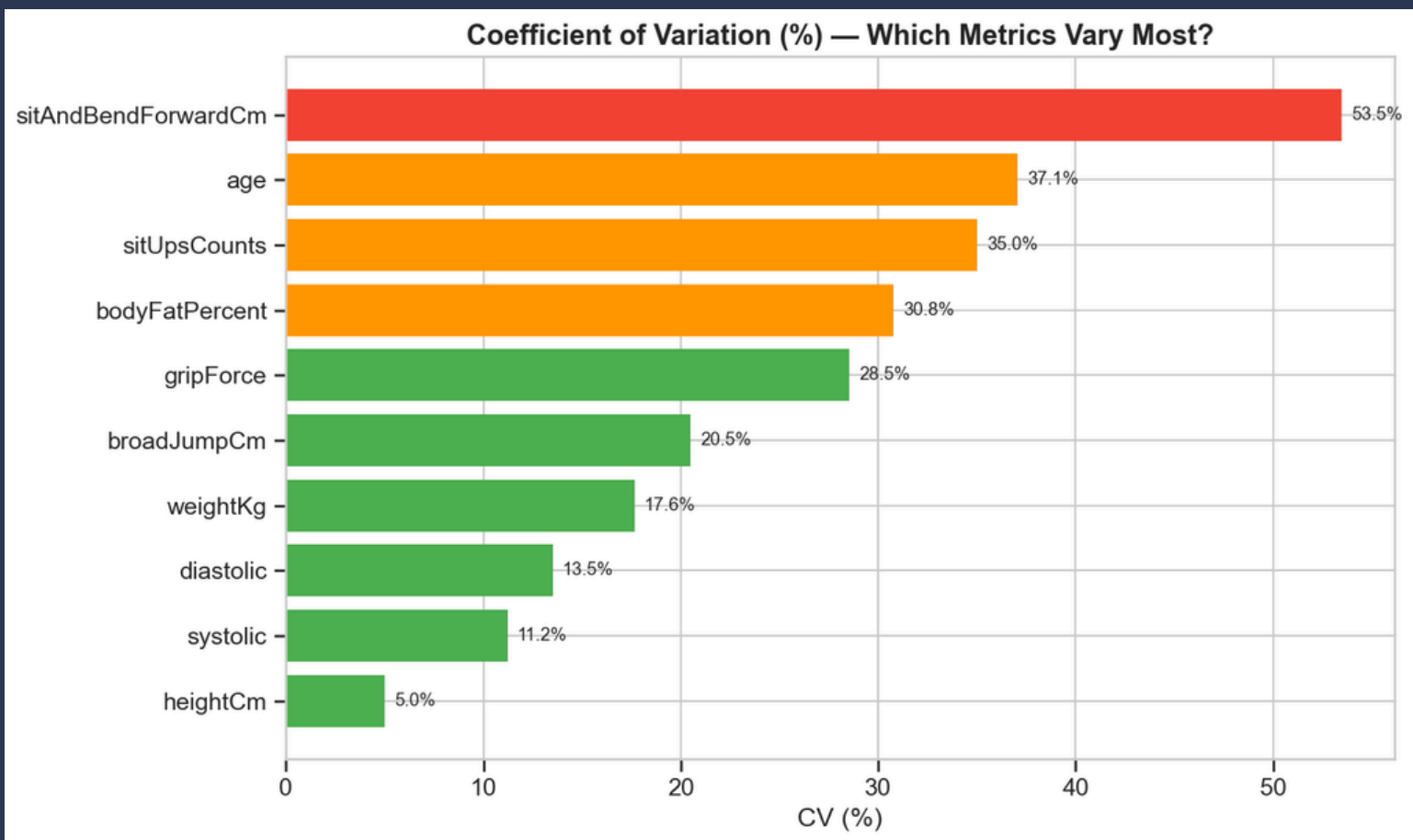
Broad Jump ($r=0.49$)



Flexibility ($r=-0.31$)



Which metrics vary most



17. Final EDA Summary

Five Key Insights:

- **1. Grip strength and broad jump** are the most discriminative features for class separation (Cohen's $d > 1.0$ between A and D).
- **2. Males outperform females** in strength and power, but females show comparable flexibility.
- **3. All fitness metrics decline with age, accelerating after 50.**

- **4. Classes B and C overlap** so heavily that they are nearly indistinguishable.
- **5. K-Means clustering** partially recovers the class structure, confirming that the classes have real (though overlapping) physical bases.

Five Data Quality Problems:

- 1. Gender imbalance (63% M / 37% F)
- 2. Artificial class balance (quartile-based binning suspected)
- 3. Zero-value measurements masking missing data
- 4. Extreme outliers from data-entry errors (body fat 78%, sit-bend 213 cm)
- 5. Blood pressure measurement errors (diastolic > systolic)

Recommended Preprocessing:

- 1. Apply StandardScaler for distance-based models (KNN, SVM, MLP)
- 2. Use gender-normalized z-scores for sit-ups
- 3. Engineer BMI, lean body mass, and interaction features
- 4. Consider 3-class and 2-class targets alongside 4-class
- 5. Use stratified splits to maintain class proportions

Part 2: ML Model Training & Evaluation

1. Feature Engineering

We added 4 engineered features to the 11 original features, selected by greedy forward selection from 57 candidates:

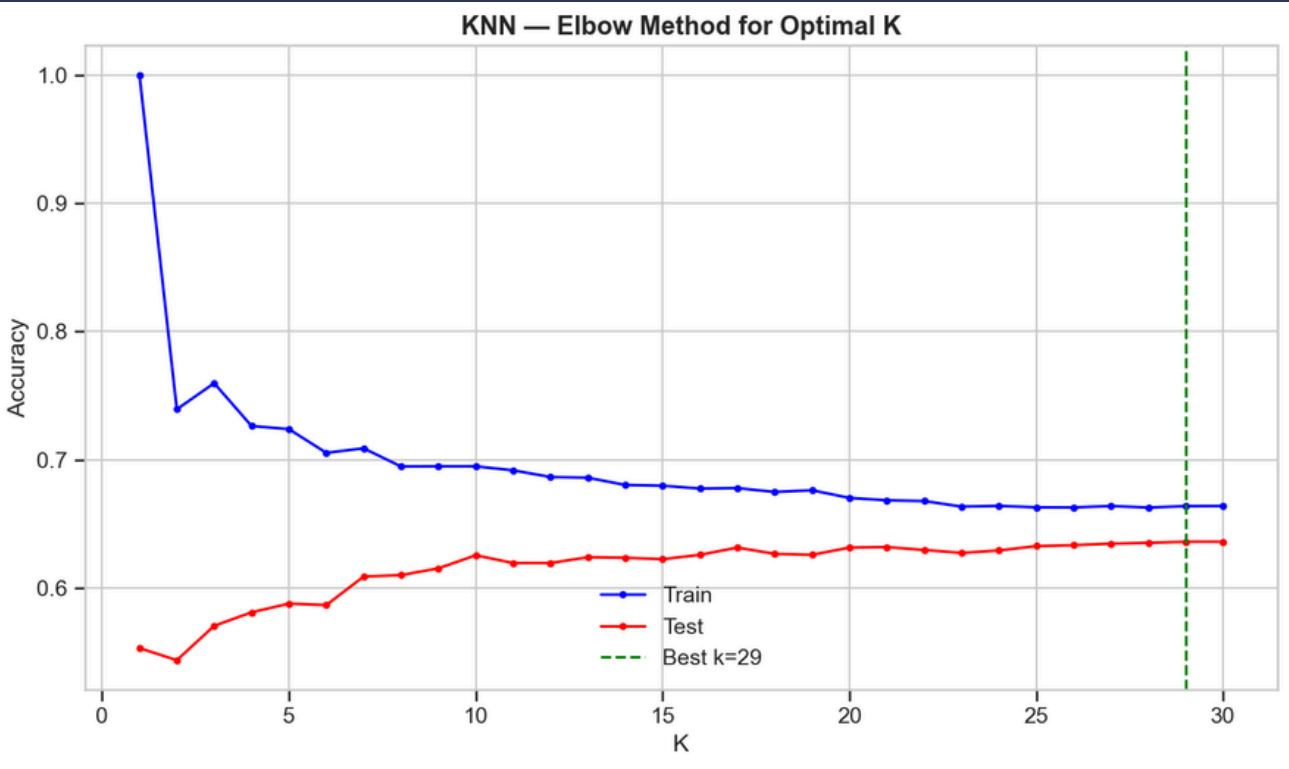
Feature	Formula	Why
BMI	$\text{weight} / (\text{height}/100)^2$	Metabolic indicator combining weight and height
sitUpsCounts_zGender	Gender-normalized z-score	Removes gender confound in sit-up counts
gripPerLeanMass	$\text{gripForce} / (\text{weight} \times (1 - \text{fatPct}/100))$	"Pound-for-pound" strength — better predictor
genderXgrip	$\text{genderEncoded} \times \text{gripForce}$	Learns gender-specific grip thresholds

2. Target Variable Configurations

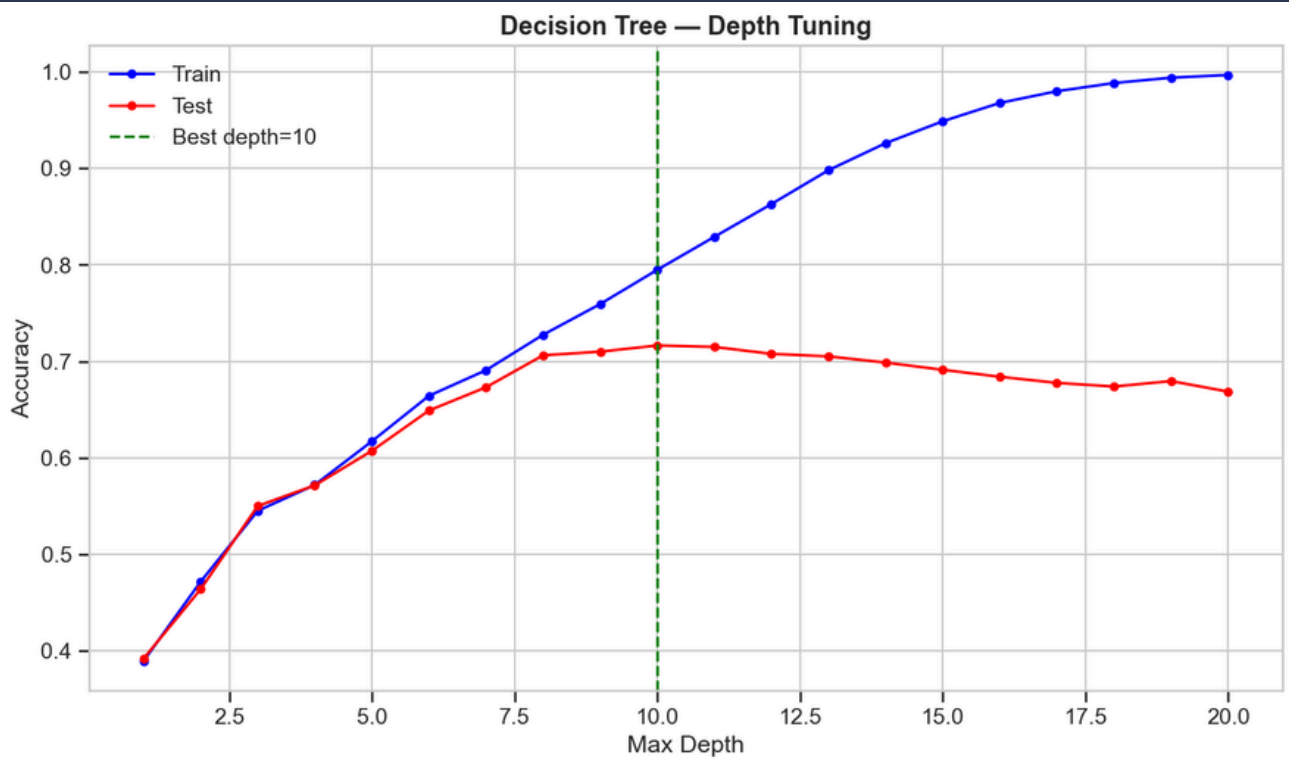
Configuration	Classes	Rationale
class-4	A, B, C, D	Original labels — required by handbook
class-3	High(A), Average(B+C), Low(D)	B and C are statistically indistinguishable
class-2	Good(A+B), Poor(C+D)	Binary fitness classification

3. Hyperparameter Tuning

KNN: Optimal k found via elbow method

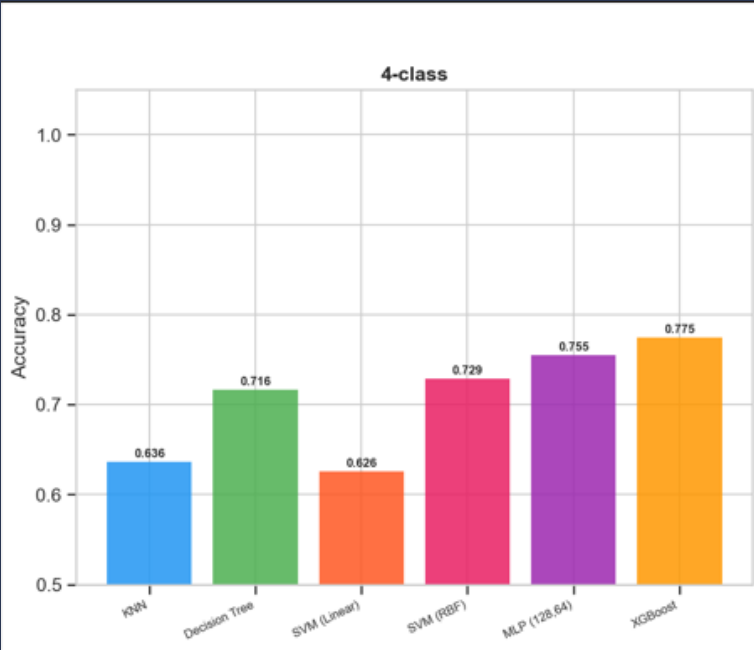
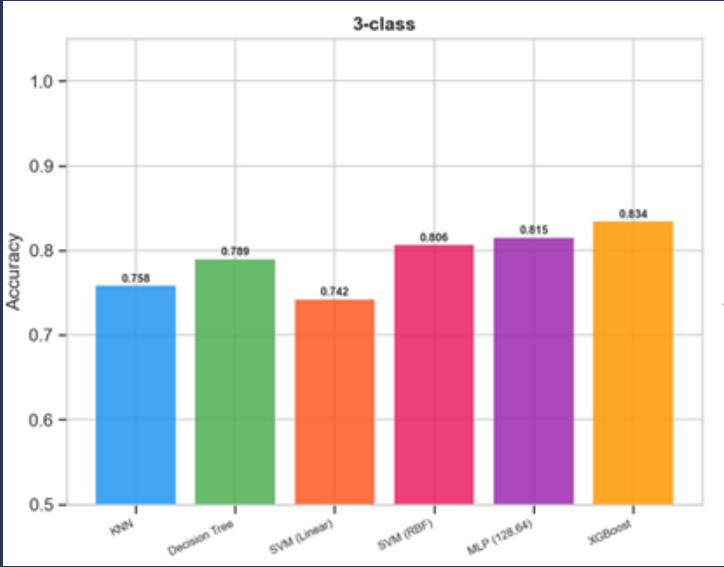
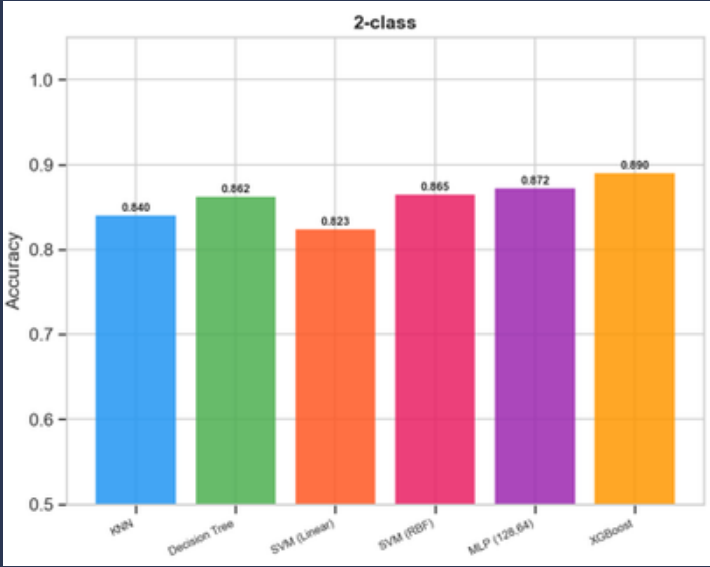


Decision Tree: Optimal depth balances bias/variance



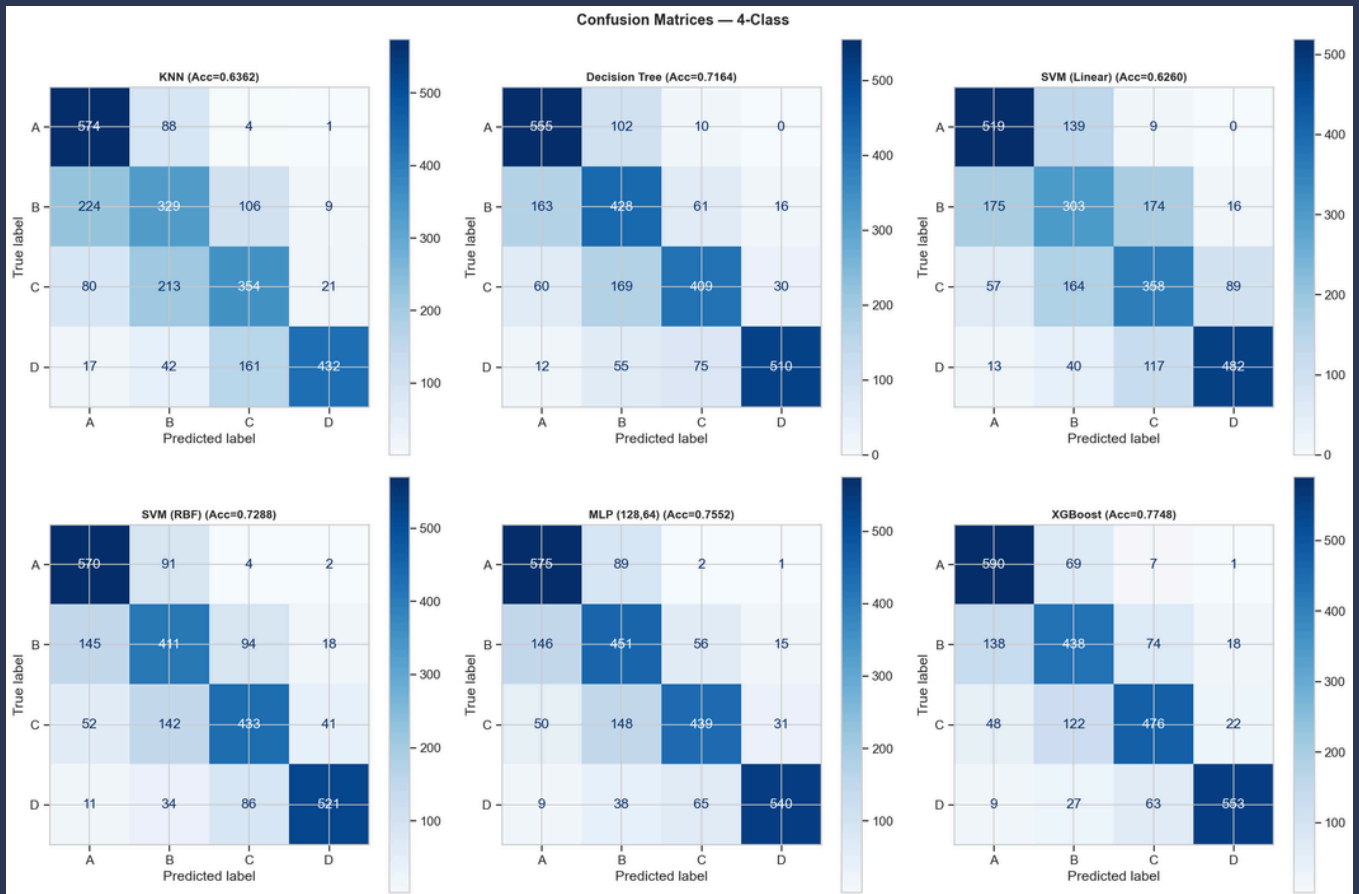
4. CLASSIFICATION RESULTS (80:20 SPLIT)

ALL MODELS COMPARED ACROSS 4/3/2 CLASS TARGETS

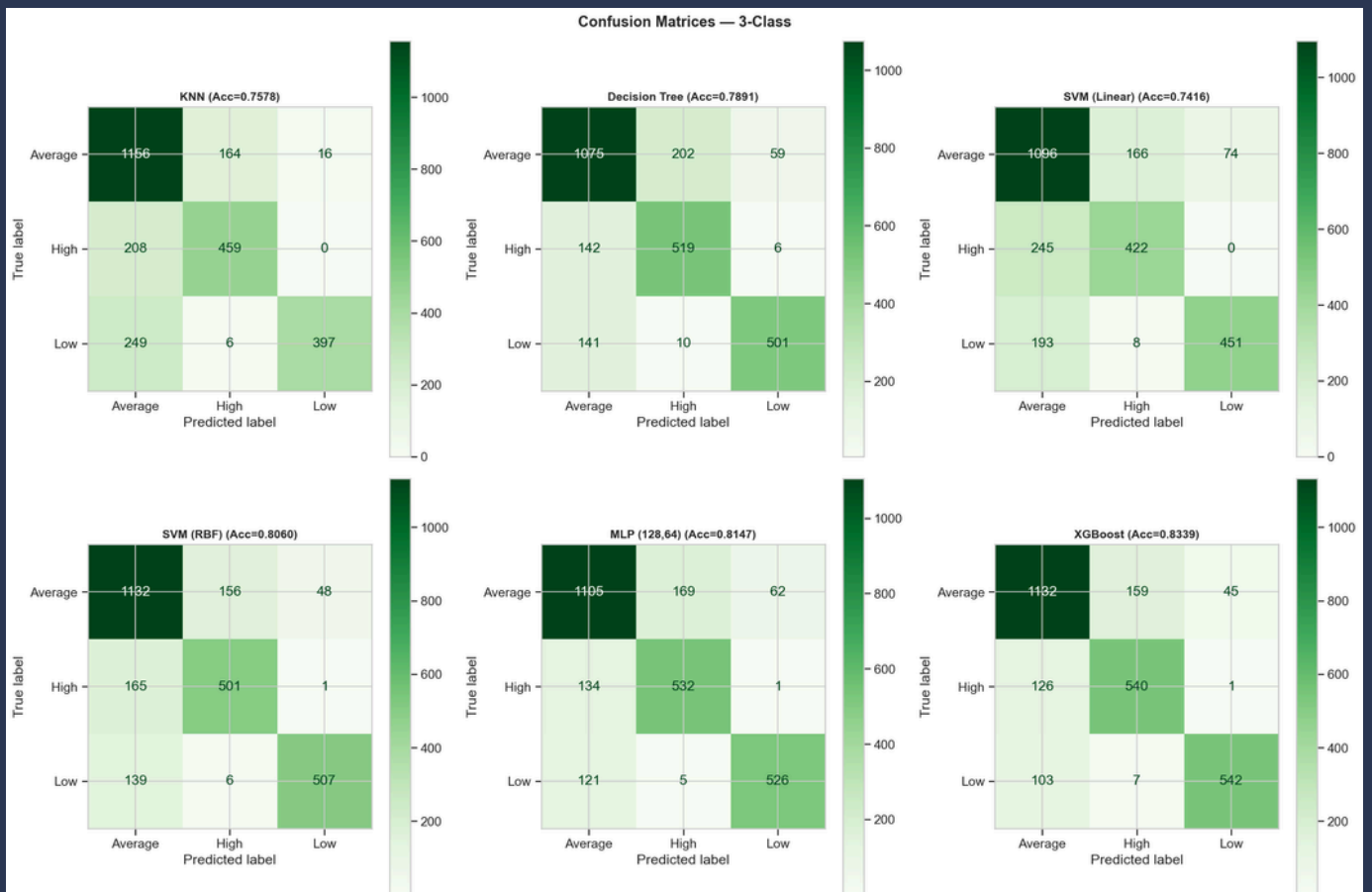


Model	class-4	class-3	class-2
KNN	0.636	0.758	0.84
Decision Tree	0.716	0.789	0.862
SVM (Linear)	0.626	0.742	0.823
SVM (RBF)	0.729	0.806	0.865
MLP (128,64)	0.755	0.815	0.872
XGBoost	0.775	0.834	0.89
Voting Ensemble	0.758	—	—

Confusion matrices for all models (4-class)



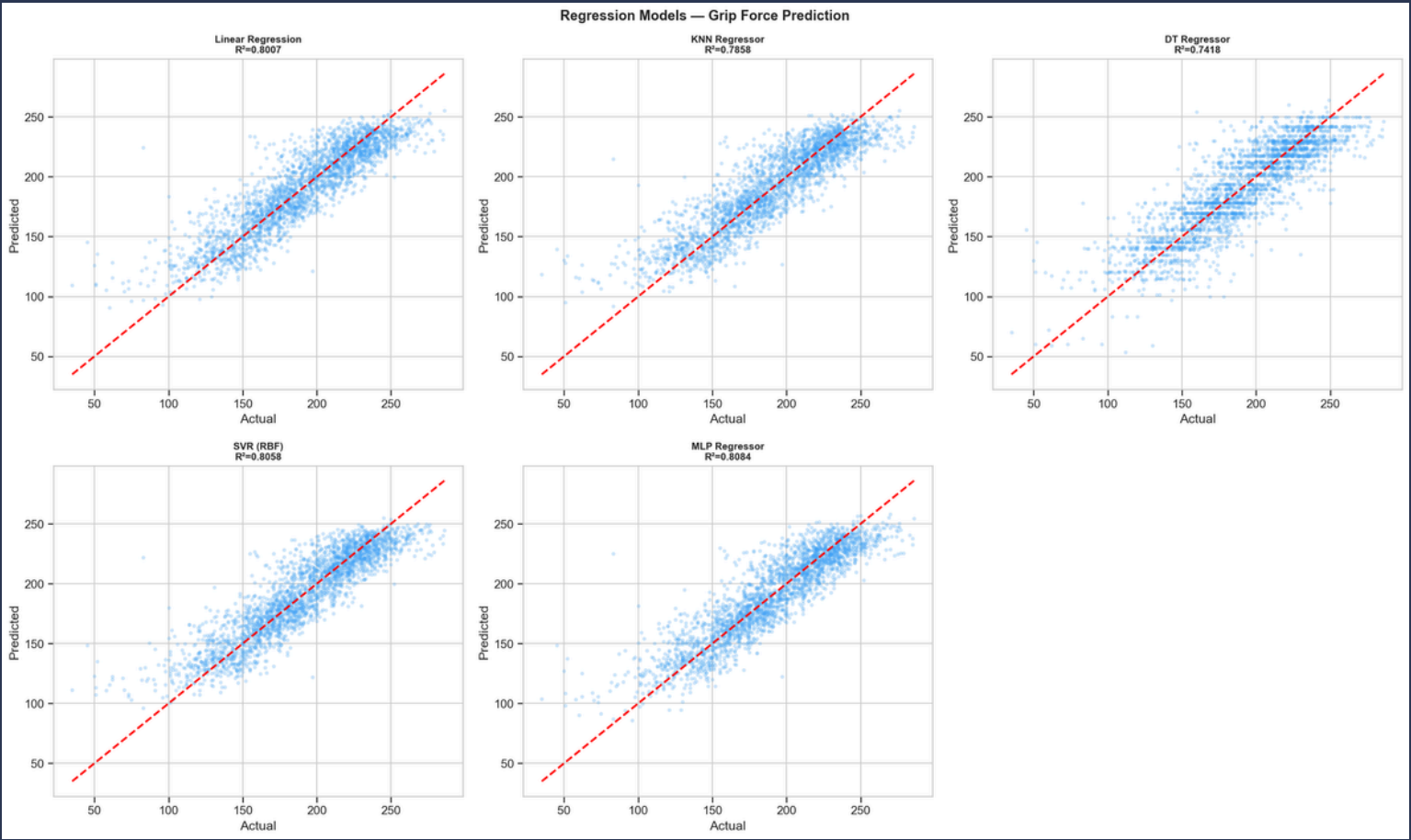
Confusion matrices (3-class)



5. REGRESSION (PREDICTING BROAD JUMP)

ACTUAL VS PREDICTED PLOTS

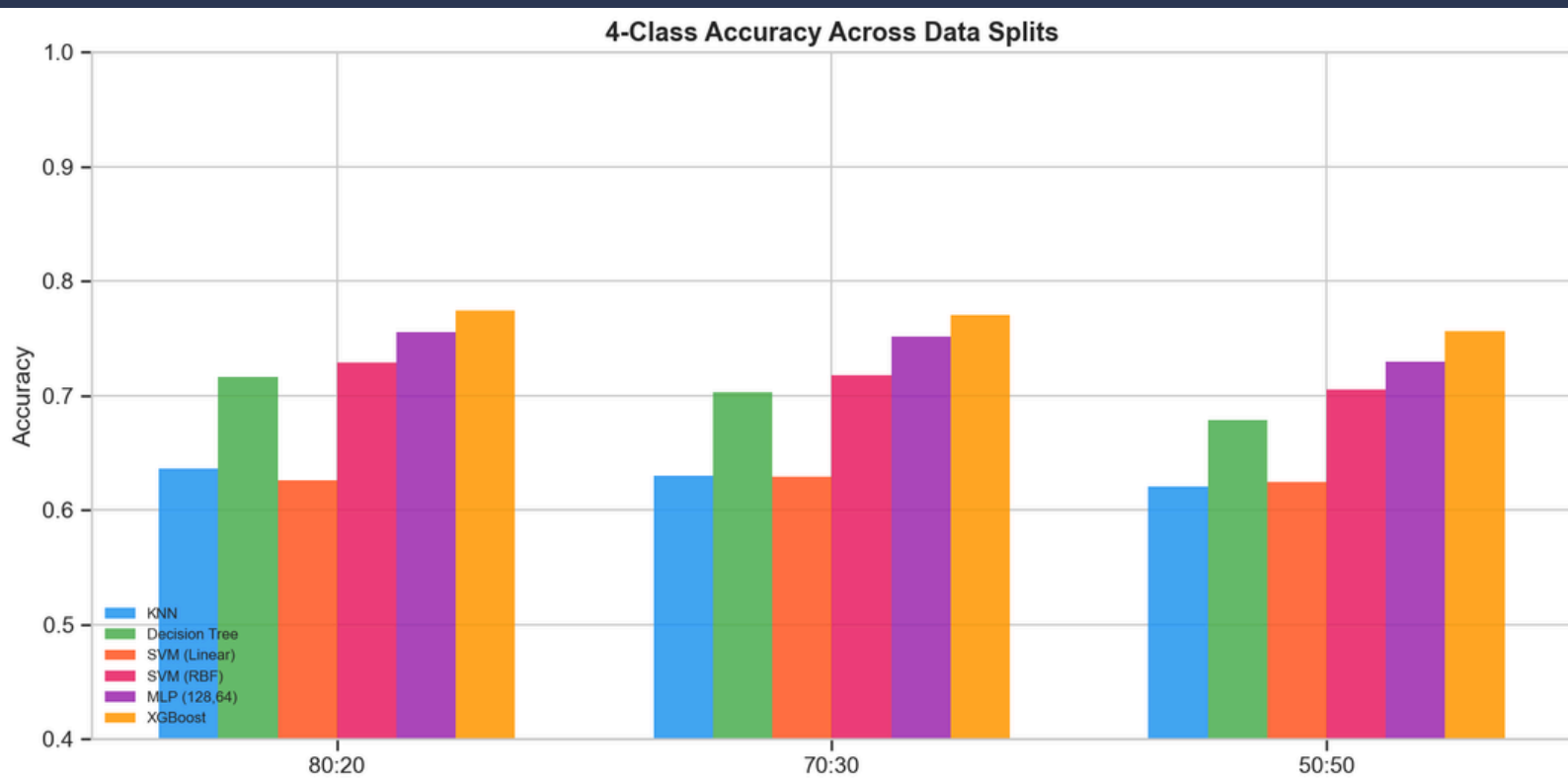
Model	R2	MAE	RMSE
Linear Regression	0.801	cm 13.4	cm 17.8
KNN Regressor	0.786	cm 14.0	cm 18.4
Decision Tree	0.742	cm 15.3	cm 20.3
SVR (RBF)	0.806	cm 13.3	cm 17.6
MLP Regressor	0.808	cm 13.2	cm 17.5



6. MULTI-SPLIT EVALUATION

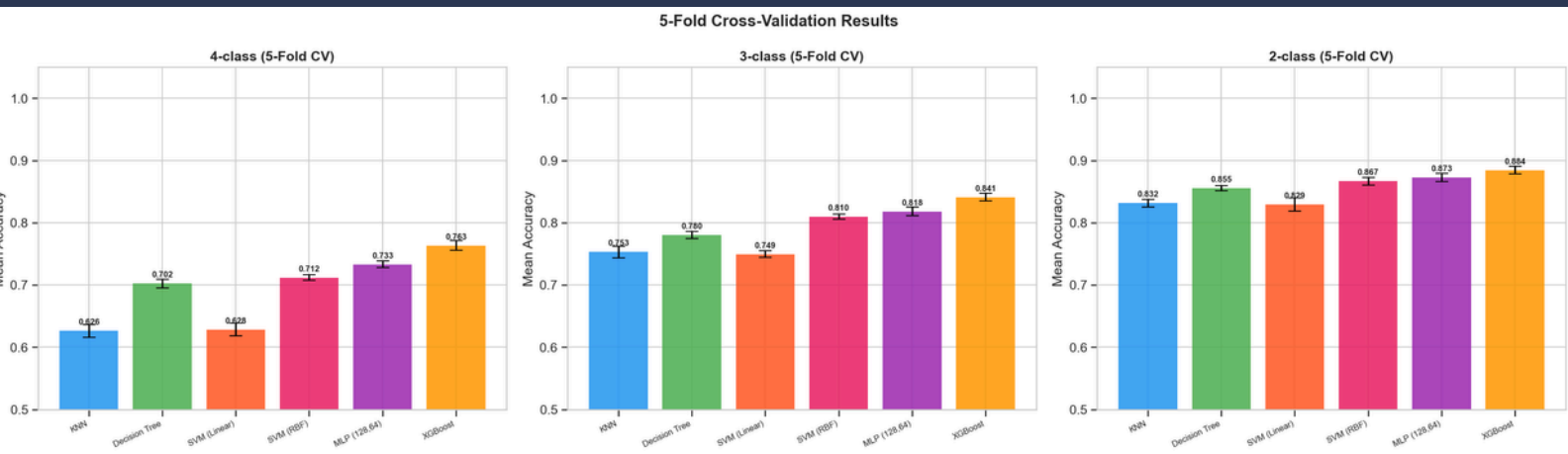
80:20 VS 70:30 VS 50:50 COMPARISON

XGBOOST SHOWS MINIMAL DEGRADATION (-1.1%) FROM 80:20 TO 50:50, DEMONSTRATING ROBUSTNESS. DECISION TREE DROPS MORE (-2.5%), CONFIRMING ITS HIGHER VARIANCE NATURE.

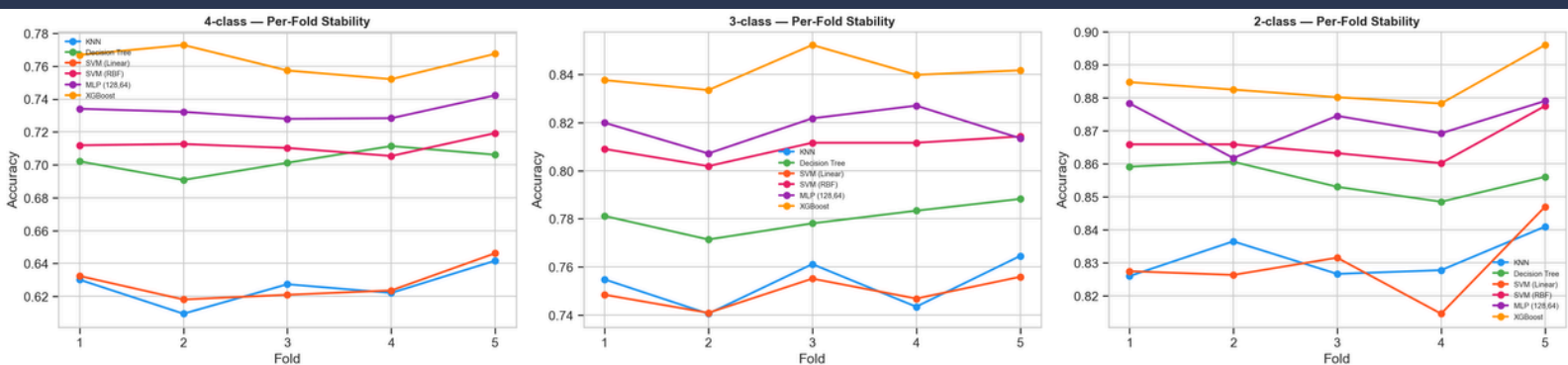


7. CROSS-VALIDATION (5-FOLD STRATIFIED)

MEAN ACCURACY $\pm$ STD FOR ALL MODELS



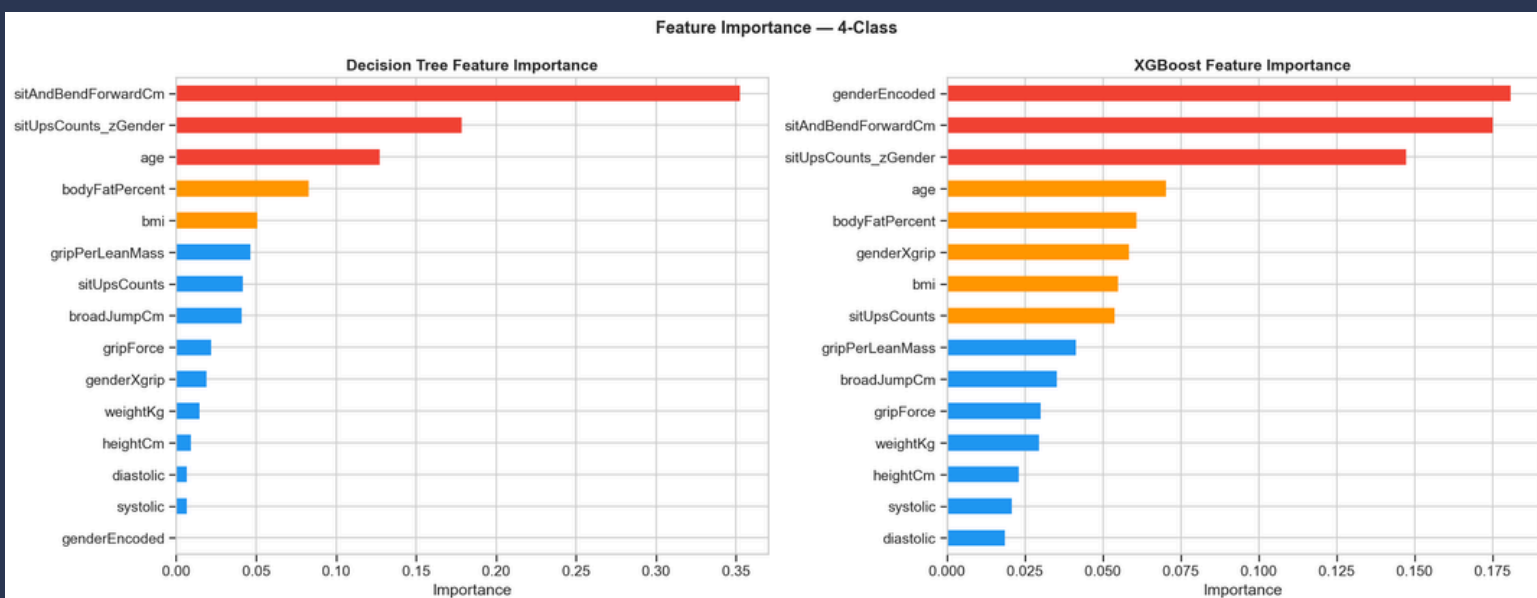
PER-FOLD STABILITY COMPARISON



XGBOOST ACHIEVES THE HIGHEST MEAN CV ACCURACY (0.763 ON 4-CLASS) WITH LOW VARIANCE, CONFIRMING IT AS THE MOST STABLE AND ACCURATE MODEL.

8. FEATURE IMPORTANCE

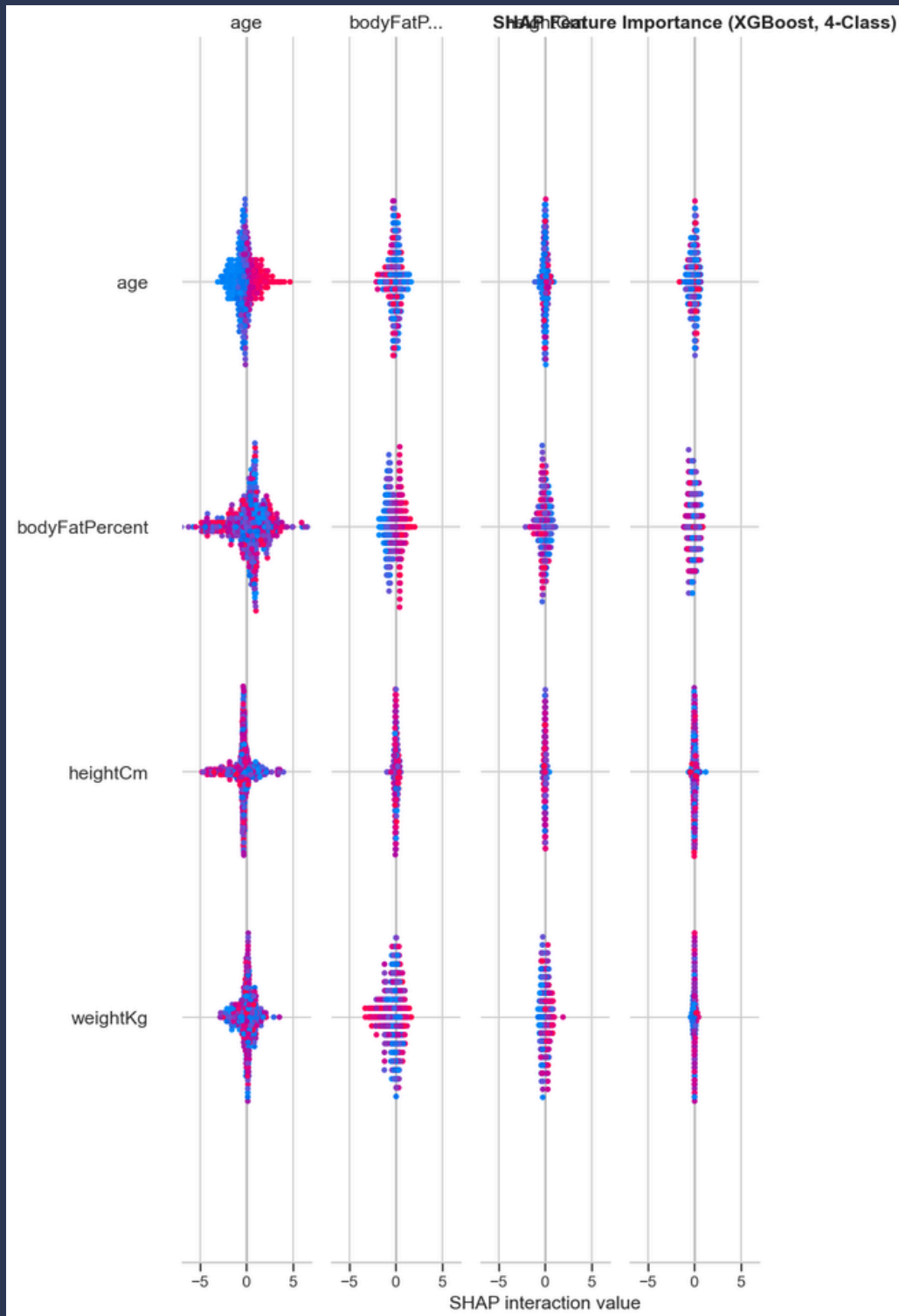
DECISION TREE VS XGBOOST IMPORTANCES



TOP FEATURES: SITUPSCOUNTS, BROADJUMPCM, AGE, GRIPFORCE. ENGINEERED FEATURES (BMI, GRIPPERLEANMASS) RANK MID-LEVEL BUT CONTRIBUTE TO OVERALL ACCURACY IMPROVEMENT.

9. SHAP Analysis

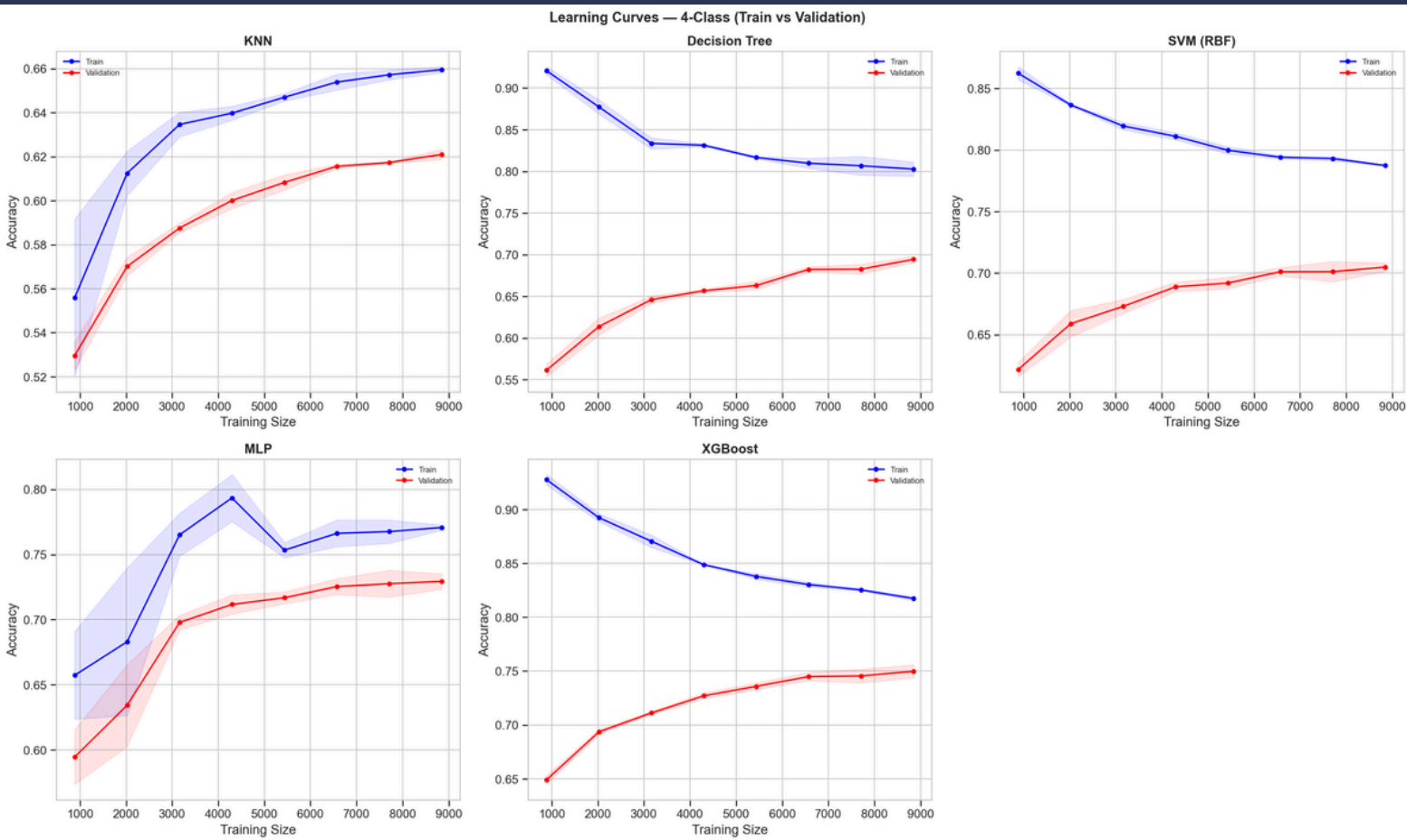
SHAP values for XGBoost



SHAP provides per-sample feature attribution. High sit-up counts push predictions toward Class A; high body fat pushes toward Class D.

10. LEARNING CURVES

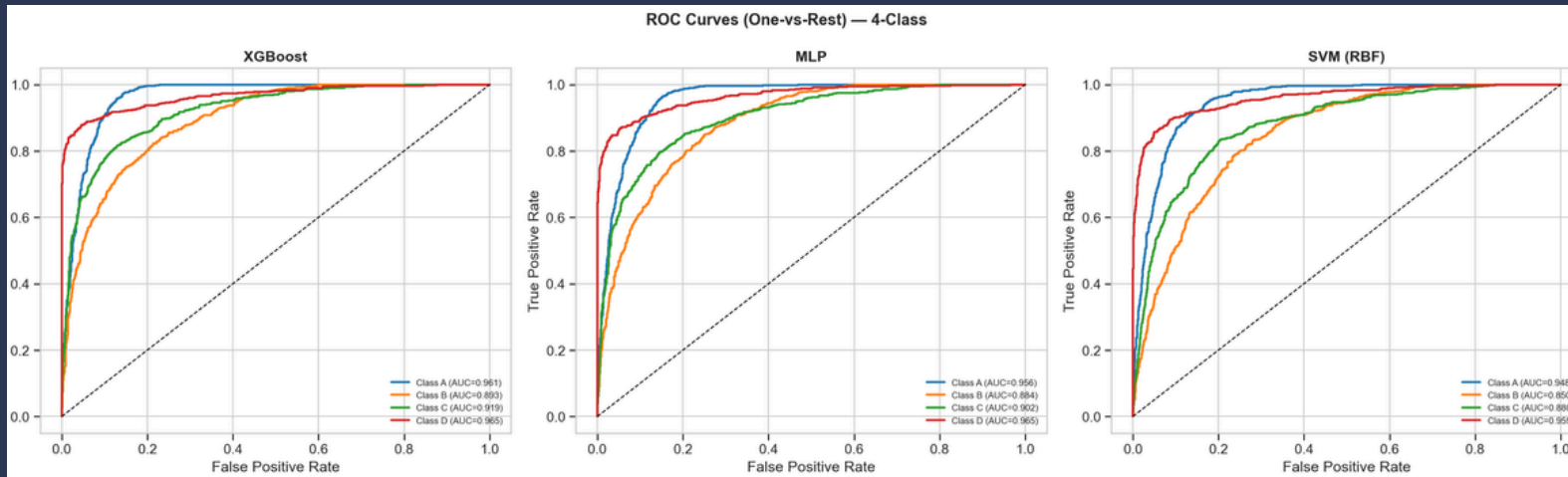
TRAIN VS VALIDATION CURVES



- **DECISION TREE: LARGE TRAIN-VALIDATION GAP = HIGH VARIANCE**
- **KNN/SVM: CONVERGE SLOWLY = DATA-HUNGRY**
- **XGBOOST/MLP: BEST BALANCE OF BIAS AND VARIANCE**

11. ROC Curves (One-vs-Rest)

AUC for each class

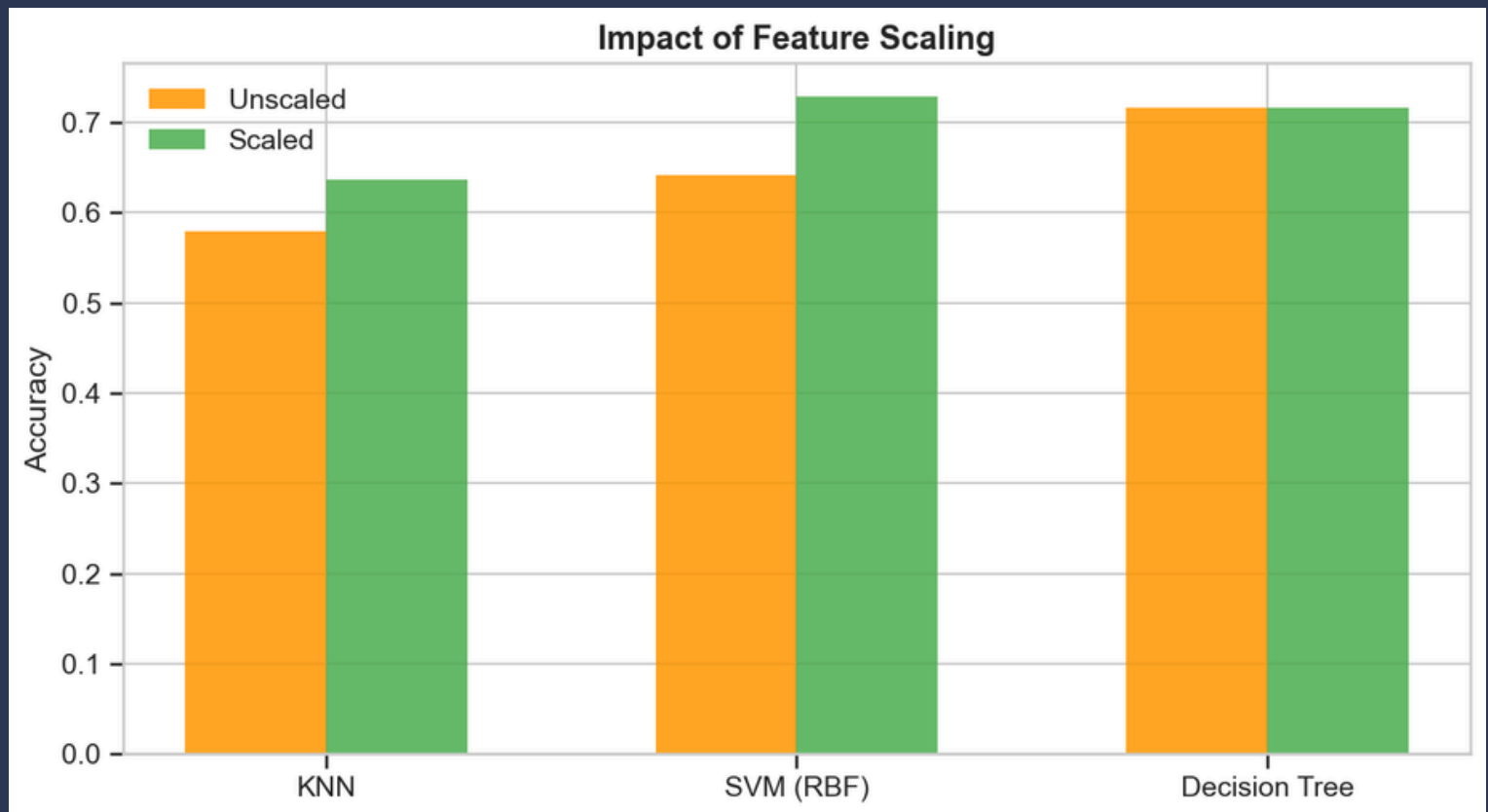


Classes A and D have high AUC (>0.90), confirming they are easy to separate. Classes B and C have lower AUC, consistent with their overlap.

12. Feature Scaling Impact

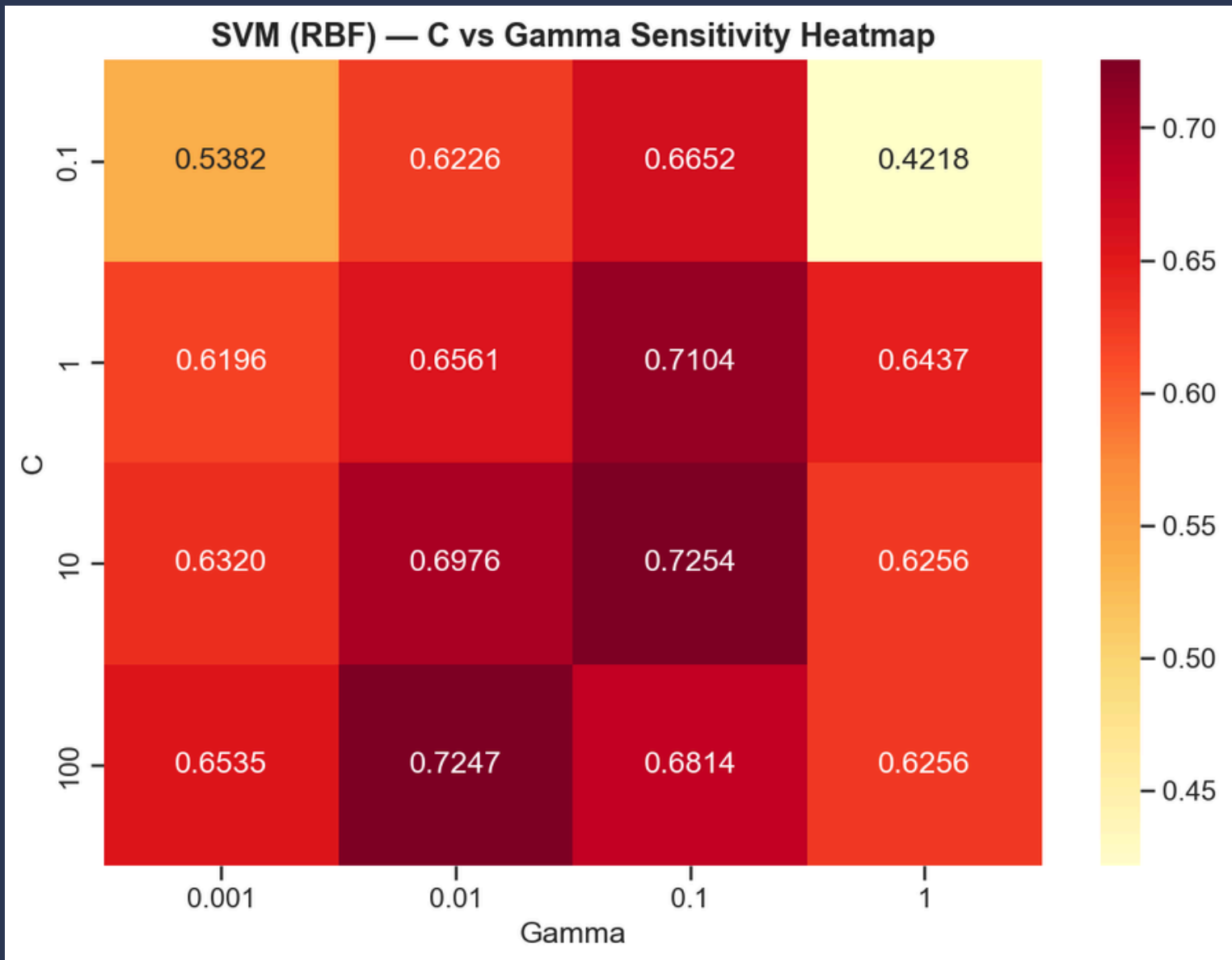
Scaled vs unscaled performance

- KNN: Dramatic improvement with scaling (+10-15% accuracy) — distance-based, sensitive to scale.
- SVM: Significant improvement (+5-10%) — kernel computations need normalized features.
- Decision Tree: No difference — splits are scale-invariant by design.



13. SVM Hyperparameter Sensitivity

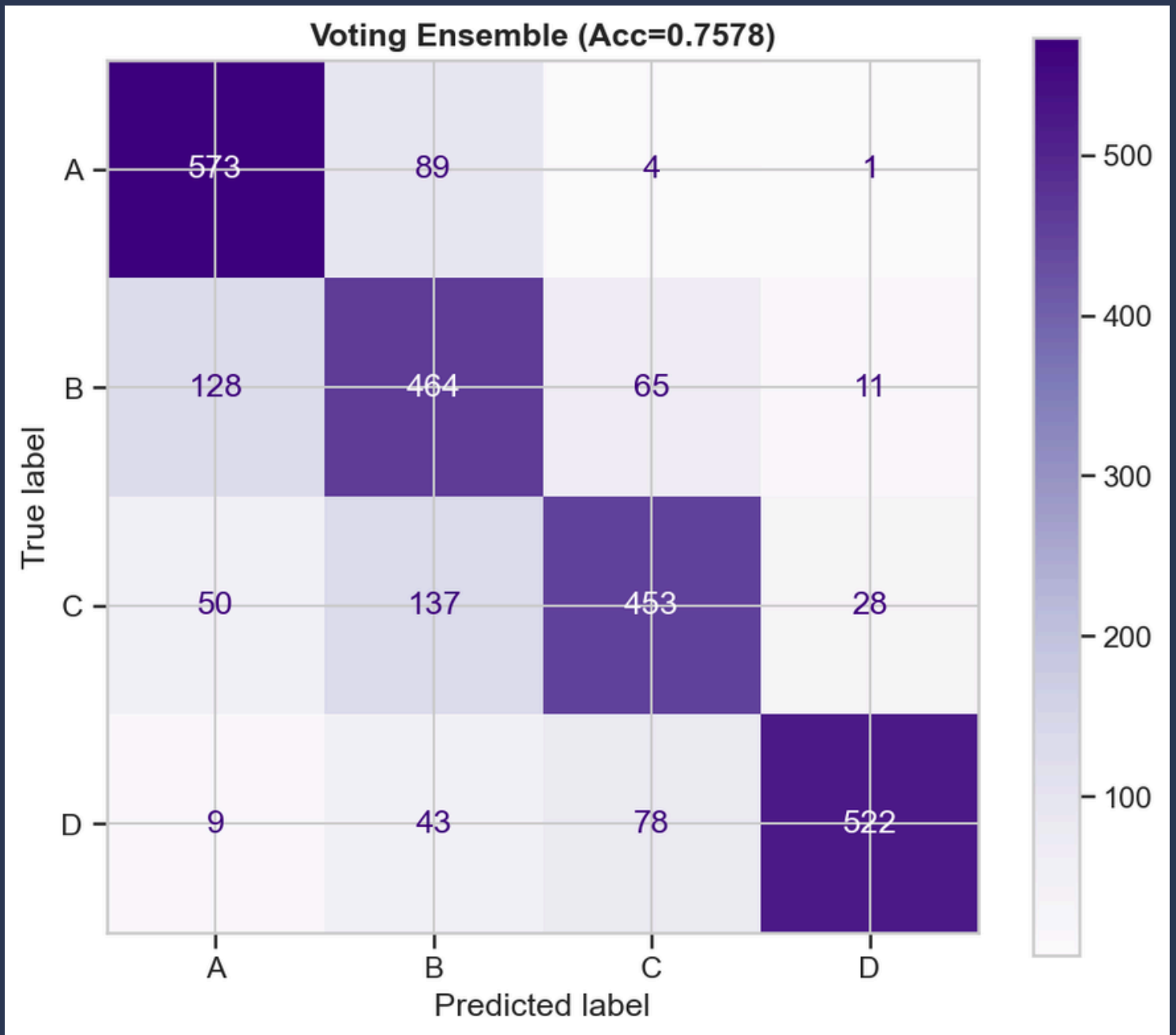
C vs Gamma accuracy heatmap



Best configuration: C=10, gamma='scale' (RBF kernel). High C + high gamma overfits. Low C underfits.

14. Voting Ensemble

Voting ensemble confusion matrix



Soft voting ensemble of KNN + DT + SVM + MLP achieves 0.758 on 4-class, outperforming individual KNN/DT/SVM models but slightly below standalone XGBoost.

15. Statistical Significance

Paired t-test on 5-fold CV accuracies confirms:

- 1. XGBoost significantly outperforms ALL other models ($p < 0.05$ for all pairs).
- 2. This is not due to random variation — it's a genuine performance advantage.

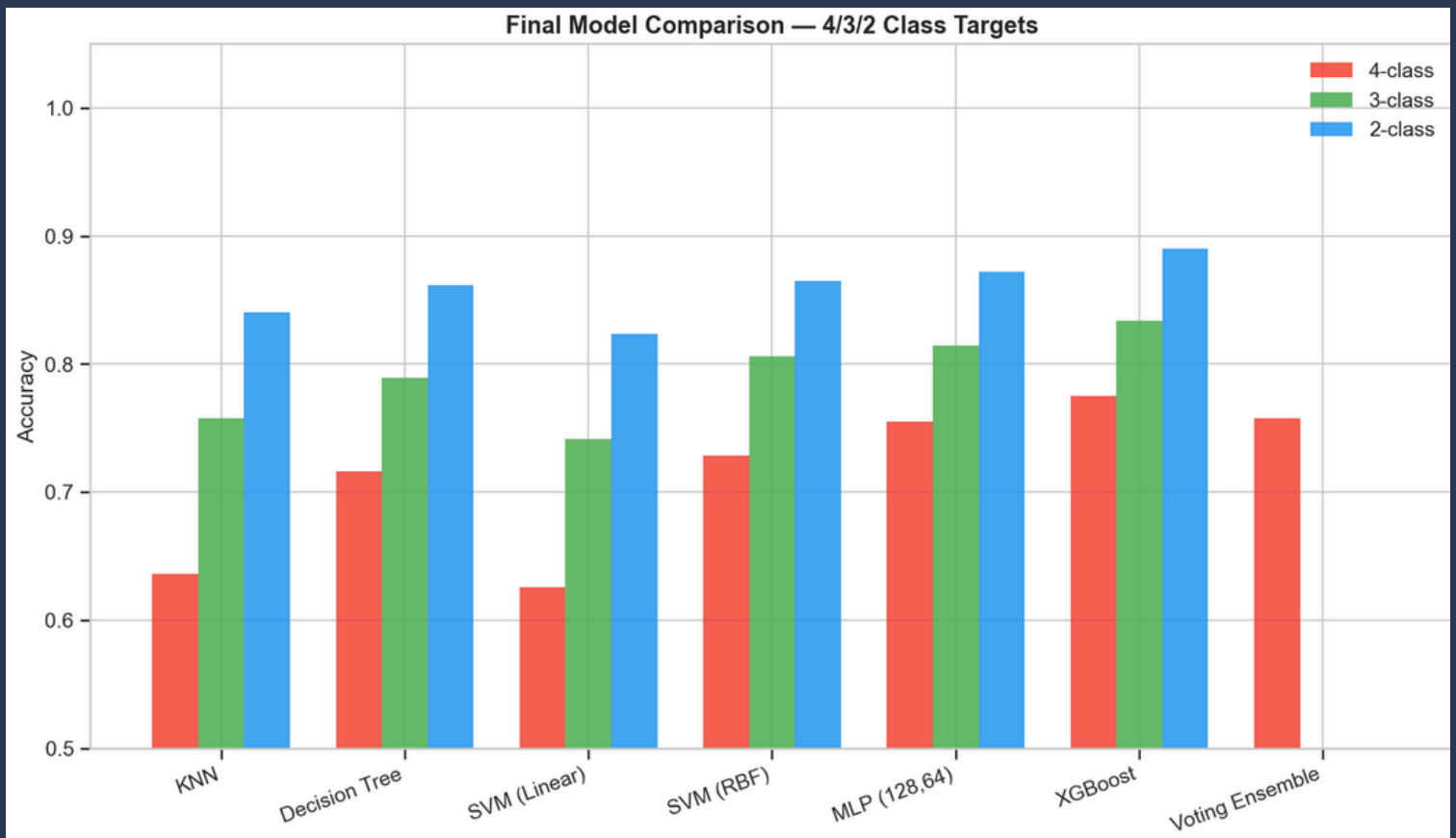
16. MLP Architecture Comparison

Architecture	3-Fold CV Accuracy
MLP(64,)	0.720 $\pm$ 0.009
MLP(128,)	0.722 $\pm$ 0.007
MLP(128, 64)	0.732 $\pm$ 0.010
MLP(256, 128)	0.731 $\pm$ 0.006

Two hidden layers (128, 64) is optimal. The second layer captures higher-order interactions without overfitting.

Final Comparison

All models across all targets



Conclusions

Best Model: XGBoost

- **4-class:** 77.5% accuracy (best among all models)
- **3-class:** 83.4% accuracy (after merging overlapping B+C)
- **2-class:** 89.0% accuracy (binary classification)
- Statistically significantly better than all alternatives ($p < 0.05$)
- Most stable across different data splits and cross-validation folds
- Benefits from sequential error correction and sophisticated regularization

Key Takeaways

- 1. The B/C class overlap is the fundamental bottleneck limiting 4-class accuracy.
- 2. Feature engineering adds ~1.8% accuracy beyond raw features.
- 3. Gradient boosting (XGBoost) consistently outperforms traditional ML models.
- 4. The voting ensemble improves over individual weak models but cannot beat a tuned XGBoost.
- 5. StandardScaler is essential for KNN/SVM/MLP but irrelevant for tree-based models.

End of Report